

Design and Development of a Hydrodynamic Tunnel for Experimental Fluid Dynamics

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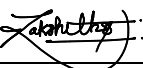
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Abstract

Hydrodynamic tunnels are essential experimental facilities for investigating fluid flow behavior around submerged bodies under controlled laboratory conditions, with applications in marine engineering, hydrofoil testing, high Reynolds number testing, and CFD model validation. This project presents the design and development of a small/medium-scale, vertical, closed-loop hydrodynamic tunnel with an open-channel test section, intended to produce steady, uniform, low-turbulence flow for external flow experiments. The primary experimental objective is to reproduce and observe distinct flow regimes around a cylindrical body, from laminar steady flow through to transitional unsteady flow, corresponding to a Reynolds number range of 0 to approximately 8,000, based on cylinder diameter as the characteristic length.

The design process involved a systematic literature-based configuration selection, followed by analytical calculations for all major components including the contraction section, wide-angle diffuser, vertical diffusers, settling chamber, honeycomb structure, and 90° bends. A MATLAB-based head loss model was developed and confirmed that total system head loss (~0.9 m) is well within the available pump head (~9 m), indicating that head loss alone is insufficient as an optimization criterion. Component optimization was therefore carried out using single-phase ANSYS Fluent CFD simulations. CFD analysis of the wide-angle diffuser revealed significant flow separation and turbulence, which was resolved by introducing internal guide vanes, reducing the effective diffusion angle to 7° per channel. Finite Element Analysis (FEA) of glass fiber components identified that wall thickness alone was insufficient to withstand internal water column pressure, leading to the addition of ribs and reinforcements across all fabricated parts.

All primary structural components including two vertical diffusers, two 90° bends, the wide-angle diffuser with integrated vanes, the settling chamber, the contraction section, the test section, and the honeycomb structure were fabricated using glass fiber. Hydrogen bubble flow visualization was integrated into the test section to enable qualitative observation of flow patterns during experimental runs.

Keywords: hydrodynamic tunnel, flow conditioning, CFD result validation, glass fiber fabrication, Reynolds number

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List of Abbreviations

HDT	Hydrodynamic tunnel
WAD	Wide angle diffuser
AR	Area Ratio
BC	Boundary Condition
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CNC	Computer Numerical Control
C _p	Pressure Recovery Coefficient
DC	Direct Current
DNS	Direct Numerical Simulation
FEA	Finite Element Analysis
FoS	Factor of Safety
GFRP	Glass Fibre Reinforced Polymer
GSM	Grams per Square Metre
HDT	Hydrodynamic Tunnel
IEEE	Institute of Electrical and Electronics Engineers
L/D	Length-to-Diameter Ratio
L/W	Length-to-Width Ratio
LCC	Large Cavitation Channel
LDV	Laser Doppler Velocimetry
LED	Light-Emitting Diode
LES	Large Eddy Simulation
MATLAB	Matrix Laboratory
MTS	Millimetre-Tonne-Second (unit system)
NASA	National Aeronautics and Space Administration
PIV	Particle Image Velocimetry
PISO	Pressure-Implicit with Splitting of Operators
RANS	Reynolds-Averaged Navier-Stokes
Re	Reynolds Number
SST	Shear Stress Transport
uPVC	Unplasticized Polyvinyl Chloride
UGA	University of Georgia
WAD	Wide-Angle Diffuser

1 Chapter 1 - Introduction

1.1 Background and Motivation

Experimental fluid dynamics plays a fundamental role in advancing the understanding of fluid flow behavior, providing empirical data that theoretical models and numerical simulations alone cannot fully capture. Among the experimental facilities used for this purpose, hydrodynamic tunnels (HDT) occupy a particularly important position. By circulating water through a controlled test environment, these tunnels allow researchers to study fluid flow around submerged and immersed bodies under repeatable, well-defined conditions. Their applications span a wide range of engineering disciplines, including marine vessel design, underwater vehicle hydrodynamics, propeller and hydrofoil performance evaluation, cavitation studies, and the characterization of boundary layer behavior. Beyond their direct experimental value, HDTs serve as essential tools for validating Computational Fluid Dynamics (CFD) models, providing a physical reference against which numerical predictions can be tested and refined.

A key advantage of water as a working fluid is its high density and relatively low kinematic viscosity compared to air, which allows meaningful Reynolds numbers to be achieved at significantly lower flow velocities than would be required in a wind tunnel. This makes HDTs particularly well-suited to studying external flow phenomena such as vortex shedding, flow separation, and boundary layer transition, at laboratory scale, without the need for large infrastructure or high-power drive systems.

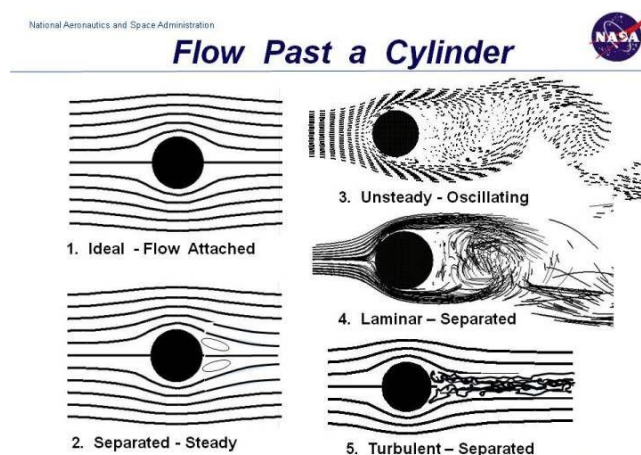


Figure 1.1: flow pattern progression over circular cylinder
Source: [Drag of a Sphere](#) | [Glenn Research Center](#) | [NASA](#)

For a canonical experiment such as flow over a circular cylinder, a HDT can reproduce the full progression of flow regimes, from laminar steady flow at very low Reynolds numbers through the development of a steady vortex pair, the onset of the laminar vortex street, and into the transitional unsteady regime, all within a compact and manageable facility.

Despite the clear research and educational value of such facilities, access to purpose-built hydrodynamic testing infrastructure remains limited in Sri Lanka. Institutions engaged in fluid mechanics research and engineering education currently have no locally available facility for conducting controlled hydrodynamic experiments, creating a dependence on overseas testing that is both costly and logistically impractical for most academic and industry purposes. This gap is particularly significant given the growing interest in CFD-based design across local engineering sectors, where experimental validation remains a critical but largely unmet need.



Figure 1.2: Small to medium scale water tunnel
Source: [*Low Speed Water Tunnel – Experimental Flow Physics Lab*](#)

This project is motivated by the need to address this gap through the design and development of a functional, cost-effective hydrodynamic tunnel suitable for a university laboratory setting. By leveraging locally available materials, a pump already procured by the department, and fabrication methods accessible within the university's facilities, the project aims to produce a tunnel capable of generating steady, uniform, low-turbulence flow in its test section across a Reynolds number range sufficient to demonstrate the principal flow regimes around a cylindrical body. In doing so, it seeks to establish a replicable design framework that can support both academic research and undergraduate instruction in experimental fluid dynamics at the University of Moratuwa and beyond.

1.2 Problem Statement

HDTs are established tools in fluid dynamics research worldwide, yet no such facility currently exists anywhere in Sri Lanka; neither in industry nor in any academic institution. This absence entirely eliminates the possibility of conducting controlled hydrodynamic flow experiments locally, forcing any such research to depend on overseas facilities, which is both financially prohibitive and logistically impractical for most academic and engineering purposes in the Sri Lankan context.

Beyond the absence of physical infrastructure, a further gap exists in the available technical literature. While numerous research publications document the design of specific HDTs, no comprehensive, accessible design framework exists that consolidates the full design process, from configuration selection and component sizing through to optimization, structural analysis, and fabrication into a coherent step-by-step guide applicable to small/medium-scale facilities. This fragmentation of knowledge creates a significant barrier for institutions seeking to design and build their own tunnel from first principles under real-world constraints of budget, space, and locally available materials and fabrication methods.

This project addresses both gaps simultaneously: by designing and fabricating Sri Lanka's first HDT within the constraints of a university laboratory setting, and by developing a consolidated design framework that documents the complete process in sufficient detail to enable replication and further development by other institutions.

1.3 Aim and Objectives

Aim

To design and develop a hydrodynamic tunnel with standard testing flow conditions in the test section and effective flow visualization methods for experimental fluid dynamics.

Objectives

1. Identification of the existing relationships between the design characteristics of the hydrodynamic tunnels and the flow conditions within its test section.
2. Design of the hydrodynamic tunnel based on the analysis of design characteristics.
3. Optimization of the hydrodynamic tunnel design to enhance the performance and ensure the standard flow conditions.

4. Development of the hydrodynamic tunnel to achieve standard flow conditions in the test section and validation.

1.4 Scope and Limitations

This project covers the complete design, optimization, and fabrication of a small/medium-scale, vertical, closed-loop HDT with an open-channel test section, intended for use within the Department of Mechanical Engineering at the University of Moratuwa. The scope includes literature-based configuration selection, analytical calculations and MATLAB-based modelling for all major components head loss, single-phase CFD-based flow optimization, FEA-based structural analysis, glass fiber fabrication of all primary components, integration of hydrogen bubble flow visualization, and the development of a consolidated design framework for small/medium-scale hydrodynamic tunnels.

The project operates within several defined limitations. The tunnel is designed for a single fixed operating velocity rather than variable-speed operation, as flow control mechanisms were determined to be beyond the feasible scope for a three-member group within one academic year. CFD analysis is limited to single-phase simulations due to the absence of sufficient computational resources within the department; multiphase simulation of the open-channel test section, while identified as desirable, was not carried out. Fabrication is constrained to materials and methods locally available and accessible within Sri Lanka, with a total material budget of approximate

1.5 Report Structure

This report is organized into seven chapters. Chapter 2 presents a review of existing hydrodynamic tunnel designs and the theoretical background governing each major component, culminating in the justification of the selected tunnel configuration. Chapter 3 details the analytical design and calculation process for all components, including the MATLAB-based head loss model and the finalized CAD model. Chapter 4 covers the CFD and FEA simulation work carried out to optimize component geometry and verify structural integrity. Chapter 5 describes the fabrication process, materials used, and the current assembly status of the tunnel. Chapter 6 presents the experimental validation of the tunnel, including flow visualization results. Chapter 7 draws conclusions from the work completed, critically evaluates the outcomes against the stated objectives, and provides recommendations for future development.

2 Chapter 2 - Literature Review

2.1 Overview of Hydrodynamic Tunnels and Their Applications

A hydrodynamic tunnel is a recirculating or once-through facility that drives water at controlled velocities through a test section, allowing the behavior of fluid flow around submerged or immersed bodies to be studied under repeatable laboratory conditions. In principle, the facility operates similarly to a wind tunnel, but uses water as the working fluid. The higher density and lower kinematic viscosity of water relative to air means that the same Reynolds number can be achieved at a significantly lower flow velocity, making hydrodynamic tunnels particularly practical for laboratory-scale experimentation without the need for large drive systems or extensive infrastructure [2].

Hydrodynamic tunnels find application across a broad range of engineering disciplines. In marine engineering and naval architecture, they are used to evaluate drag, lift, and pressure distributions on hull forms, rudders, and appendages. Propeller and hydrofoil performance, including the onset and extent of cavitation, are commonly studied in pressure-controlled tunnel facilities [7]. In underwater vehicle development, tunnels provide the controlled flow environment necessary for characterizing hydrodynamic forces and moments on submersible bodies. Environmental and civil engineering applications include the study of flow around bridge piers, offshore structures, and tidal energy devices [15].

Beyond direct experimental measurement, hydrodynamic tunnels play a critical role in the validation of CFD models. Numerical simulations of complex flow phenomena such as vortex shedding, boundary layer separation, and turbulent wakes, require experimental benchmarks against which their predictions can be assessed. Hydrodynamic tunnels provide this benchmark by generating well-characterized, repeatable flow conditions that can be directly compared to simulation outputs [8].

At the educational level, small and medium-scale hydrodynamic tunnels serve as effective demonstration tools, allowing students to observe fundamental flow phenomena such as laminar-to-turbulent transition, the development of a von Kármán vortex street behind a cylinder, and the effects of streamlining on flow separation phenomena that are otherwise only accessible through abstract numerical or theoretical treatment [15], [16].

2.2 Tunnel Configurations - Comparative Analysis

Hydrodynamic tunnel design involves several fundamental configuration decisions, each of which significantly influences flow quality, achievable Reynolds numbers, fabrication complexity, space required to build, and cost. The three primary decisions concern tunnel orientation, circuit type, and test section arrangement.

2.2.1 Horizontal vs. Vertical Arrangements

Horizontal tunnels are by far the most common arrangement in the literature, primarily because they allow straightforward access to the test section, simplify instrumentation mounting, and present fewer structural challenges. However, they require considerable floor space, which is a significant constraint in most university laboratory settings [6].

Vertical tunnels stack components along a vertical axis, reducing the horizontal footprint considerably. An additional advantage is that the vertical orientation facilitates the natural removal of air bubbles from the circuit, improving flow quality. The primary drawbacks are increased complexity in structural design, more difficult access to components, and the added challenge of managing free-surface behavior in open-channel test sections. The Chalmers University HiCAT tunnel is a notable example of a successful vertical closed-loop design, achieving flow velocities of up to 13 m/s within a compact laboratory footprint [2].

2.2.2 Open-Circuit vs. Closed-Circuit Designs

In an open-circuit tunnel, water is drawn from a reservoir, passed through the test section once, and discharged. While simple to construct, this arrangement is wasteful of water, difficult to maintain at stable temperatures, and cannot sustain the consistent flow conditions required for precision experiments [4].

Closed-circuit designs recirculate the same body of water continuously, enabling stable, repeatable flow conditions with precise control over velocity and temperature. The enclosed nature of the circuit also reduces susceptibility to external contamination. These advantages make closed-circuit configurations the standard choice for research-grade facilities, despite their greater fabrication complexity [2], [8].

2.2.3 Open-Channel vs. Closed-Channel Test Sections

A closed-channel test section fully encloses the flow, enabling higher velocities and independent pressure control, which is essential for cavitation research. However, it demands pressure-tight construction and more complex sealing arrangements [2].

An open-channel test section incorporates a free water surface, which simplifies construction considerably by eliminating the need for pressure sealing. It also provides direct access for probe insertion and facilitates flow visualization techniques. The limitation is that free-surface wave formation constrains the maximum achievable velocity and the open-channel arrangement offers a practical and cost-effective solution [4], [5].

The classification of hydrodynamic tunnels is illustrated in Figure 2.1, which presents a hierarchical breakdown of closed-loop water tunnel systems based on scale and channel configuration.

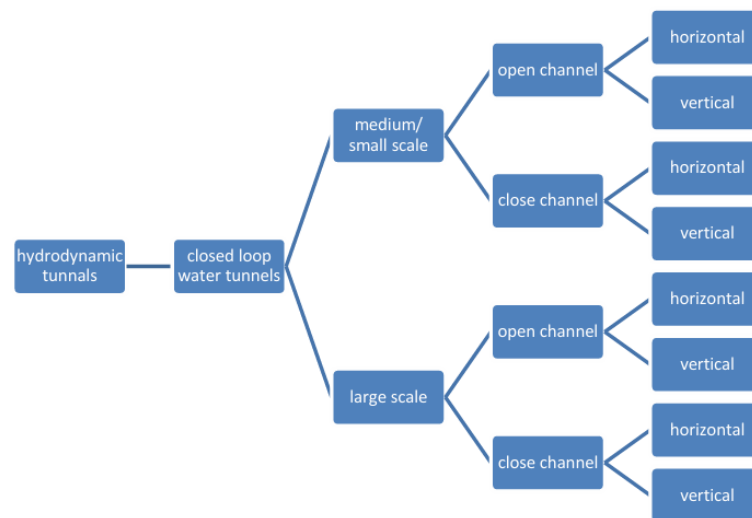


Figure 2.1: Different types of HDT configurations
Source: G14 Literature review report

The key characteristics of each configuration are summarized in Table 2.1:

Table 1: Comparison of hydrodynamic tunnel configuration options

Configuration	Max. Velocity	Flow Control	Fabrication Complexity	Space Requirement
Horizontal, closed-circuit	High	Good	Moderate	Large floor area
Vertical, closed-circuit	High	Good	High	Small floor area
Open-circuit	Low–Moderate	Poor	Low	Moderate
Open-channel test section	Low–Moderate	Moderate	Low	—
Closed-channel test section	High	Excellent	High	—

2.3 Key Design Parameters and Their Influence on Flow Quality

The flow quality within the test section of a hydrodynamic tunnel characterized primarily by velocity uniformity, turbulence intensity, and the absence of swirl or secondary flows is not determined by any single component but by the cumulative effect of design decisions made across the entire circuit. Several key parameters govern this relationship.

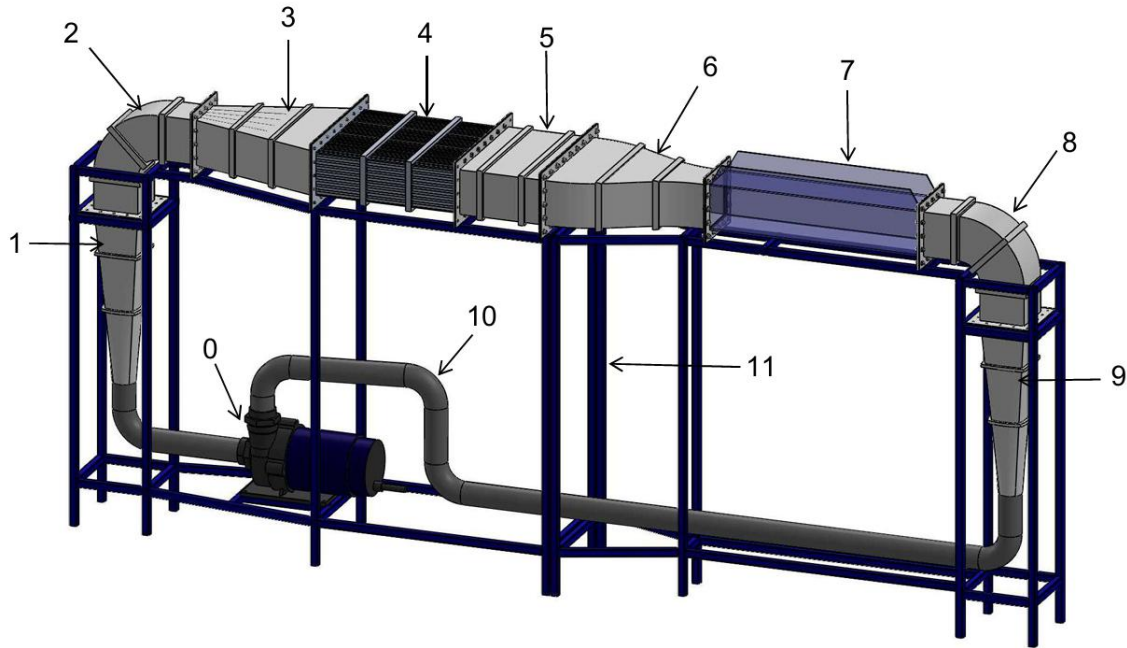


Figure 2.2: Components in selected tunnel configuration

PART NO	PART NAME
0	Water pump
1	Transition section 1
2	Rectangular bend 1
3	Wide angle diffuser
4	Honeycomb section
5	Settling chamber
6	Contraction section
7	Test section
8	Rectangular bend 2
9	Transition section 2
10	PVC pipes
11	Support structure

Contraction ratio - Contraction ratio is defined as the ratio of the inlet cross-sectional area to the test section area and is one of the most influential parameters in tunnel design. A higher contraction ratio accelerates the flow more aggressively, which suppresses upstream turbulence and non-uniformities more effectively, resulting in a more uniform velocity profile at the test section inlet. Research-grade tunnels typically employ contraction ratios

between 6:1 and 16:1, with values around 7:1 representing a practical compromise between flow quality and physical size for small/medium-scale facilities [1], [2].

Diffuser half-angle - This governs the rate at which the flow decelerates and expands after the test section. If the half-angle is too large, the adverse pressure gradient causes the boundary layer to separate from the diffuser walls, generating large-scale turbulence and flow instability that can propagate upstream. A half-angle of 7° is widely cited as the upper limit for attached flow in a conventional diffuser without additional flow conditioning [3], [9].

Honeycomb length-to-cell-diameter ratio - Honeycomb length-to-cell-diameter ratio (L/D) directly controls the effectiveness of turbulence suppression in the settling chamber. Values between 6 and 10 are recommended for most small/medium-scale tunnels, balancing turbulence reduction against pressure loss. Ratios below this range insufficiently break down large turbulent eddies, while excessively high values introduce unnecessary head loss [10].

Blockage ratio - The ratio of model frontal area to test section cross-sectional area affects the measured flow conditions around the test model. A blockage ratio below 10% is generally recommended to ensure that the presence of the model does not significantly alter the free-stream velocity [20].

Turbulence intensity - Turbulence intensity at the test section inlet is the ultimate measure of flow quality and is influenced by the combined performance of the honeycomb, screens, settling chamber length, and contraction. For most experimental purposes, a turbulence intensity below 1% is considered acceptable [1], [3].

2.4 Test Section Design and Reynolds Number Considerations

The test section is the most critical component of the hydrodynamic tunnel, as it is where experimental measurements are made and flow quality requirements are most stringent. Its design must balance several competing considerations including cross-sectional dimensions, length, shape, optical access, and structural transparency.

Cross-sectional shape and dimensions are selected based on the intended experimental objectives. Rectangular and square cross-sections are most common in small/medium-scale tunnels due to their ease of fabrication and good optical access. Three shape variants are

documented in the literature: a pure square section, a square with fixed corner fillets, and a square with longitudinally diminishing fillets.

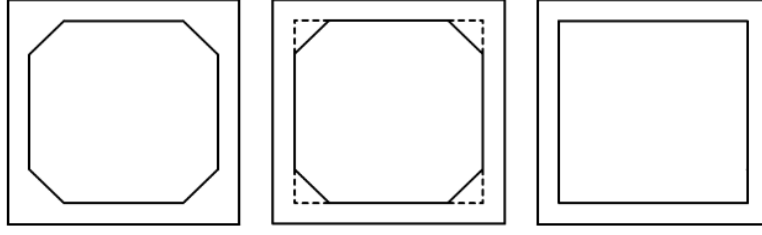


Figure 2.3: Test section shape alternatives. From left to right: square with fillets, square with diminishing fillets, and square.

The pure square maximizes optical access and is well-suited for flow visualization techniques. Corner fillets reduce secondary flows generated at sharp corners, improving velocity uniformity, while diminishing fillets additionally compensate for boundary layer growth along the test section length [2].

Test section length is governed by boundary layer development. As flow travels along the test section walls, the boundary layer thickens, progressively reducing the effective free-stream area and introducing a streamwise pressure gradient. The test section length must therefore be limited to ensure the boundary layer remains thin enough that the central flow velocity profile is not significantly affected. For the turbulent boundary layer, thickness grows approximately as,

$$\delta \propto 0.16 Re_x^{-1/7} \quad (1)$$

establishing a practical upper limit on usable length [8].

Reynolds number formulation for external flow experiments in a hydrodynamic tunnel differs fundamentally from internal flow assumptions. When testing a submerged body such as a cylinder, the relevant Reynolds number is based on the model's characteristic dimension typically its diameter rather than the hydraulic diameter of the test section. For a circular cylinder, the Reynolds number written as,

$$Re_D = \frac{UD}{\nu} \quad (2)$$

where U is the free-stream velocity, D is the cylinder diameter, and ν is the kinematic

viscosity of water. This distinction is critical because it determines the flow regime experienced by the test model independently of the tunnel dimensions [8], [3].

For the present tunnel, the target Reynolds number range of 0 to approximately 8,000 based on cylinder diameter is sufficient to reproduce the principal external flow regimes around a circular cylinder, from laminar steady flow through vortex pair formation, laminar vortex shedding, and into the transitional unsteady regime, as shown in Figure 2.2 [18].

2.5 Contraction Section Design

The contraction section accelerates the flow from the large cross-sectional area of the settling chamber to the smaller test section area, and in doing so performs three critical functions: establishing the required test section velocity, suppressing upstream turbulence and velocity non-uniformities through flow acceleration, and delivering a uniform, low-turbulence free-stream at the test section inlet [1].

2.5.1 Contraction Ratio and Velocity Uniformity

The contraction ratio, defined as the ratio of the inlet area to the exit area, is the primary design parameter governing how effectively upstream disturbances are suppressed. As the flow accelerates through the contraction, lateral velocity fluctuations are reduced proportionally to the contraction ratio, while axial fluctuations are reduced by its square root. Higher contraction ratios therefore produce more uniform and lower-turbulence flow at the test section inlet [13].

For small/medium-scale low-speed tunnels, contraction ratios between 6:1 and 10:1 are widely recommended in the literature, with a value of approximately 7:1 representing a practical design starting point that balances flow quality against the physical size of the settling chamber inlet [1], [2]. Ratios below this range provide insufficient turbulence suppression, while excessively high ratios result in impractically large inlet dimensions.

2.5.2 Contraction Curve Geometries (Bell-Mehta, Fifth-Degree Polynomial)

The shape of the contraction wall profile is equally important to the contraction ratio. An abrupt or poorly designed contraction can cause boundary layer separation on the walls, generating turbulence that directly degrades test section flow quality. Three contour types are documented in the literature: circular arc, cubic polynomial, and fifth-degree polynomial profiles.

The fifth-degree polynomial contour, originally developed by Bell and Mehta, is the most widely adopted in modern tunnel design [1], [2]. It ensures zero wall slope and zero curvature at both the inlet and exit of the contraction, providing smooth joins with the upstream settling chamber and downstream test section and eliminating inflection-point-induced separation. Cubic and circular arc contours, while simpler to manufacture, introduce inflection points that risk localized flow separation and are therefore not recommended for low-turbulence applications [13]. The Bell-Mehta polynomial profile has been successfully implemented across a wide range of tunnel scales, from small educational facilities to high-Reynolds-number research tunnels [2], [3].

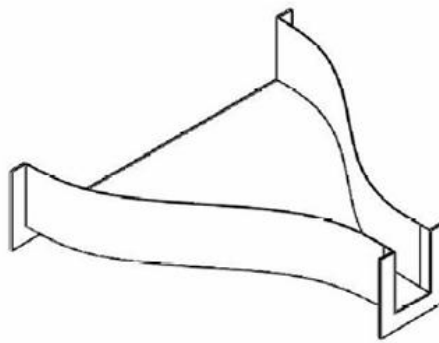


Figure 2.4: Bell-Mehta fifth-degree polynomial contraction contour

2.6 Diffuser Design

The diffuser decelerates the high-velocity flow exiting the test section, recovering static pressure and reducing the energy demand on the pump. It achieves this by gradually increasing the cross-sectional area of the flow path. The central design challenge is balancing pressure recovery against the risk of flow separation, which occurs when the adverse pressure gradient caused by area expansion exceeds the momentum of the boundary layer near the walls [3].

2.6.1 Conventional Diffusers

In a conventional straight-walled diffuser, the rate of area expansion is characterized by the half-angle of divergence. If this angle is too large, the boundary layer separates from the diffuser walls, producing large-scale turbulence, pressure fluctuations, and flow instability that can propagate upstream into the test section. A half-angle of 7° is widely established in the literature as the practical upper limit for achieving reliable pressure recovery without flow separation [3], [9].

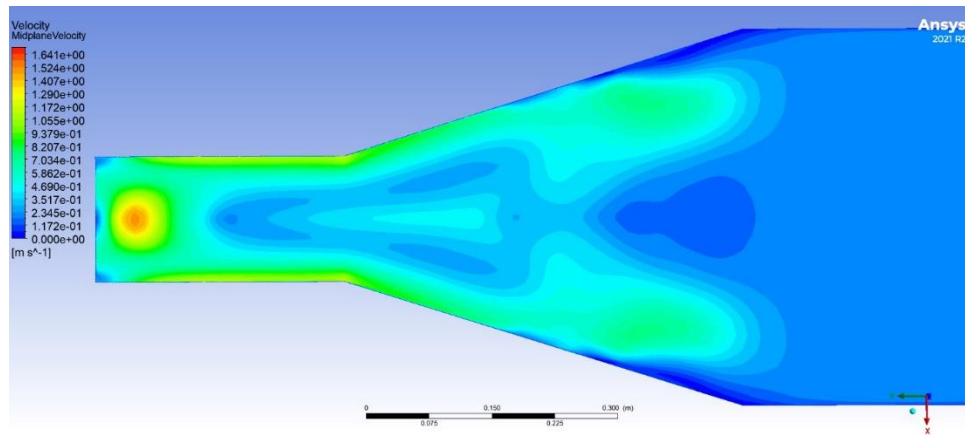


Figure 2.5: Conventional diffuser geometry showing half-angle of divergence
Source: ANSYS Flow simulation results

To meet the area ratio requirements of a tunnel circuit while maintaining this conservative angle, conventional diffusers must be made physically long. In space-constrained laboratory settings this is often impractical, and multi-stage diffusion strategies are commonly employed instead. The University of New Hampshire HiCAT tunnel, for example, uses three sequential diffusers with half-angles of 3.5° , 6° , and 4° respectively to manage the progressive deceleration of flow across the full circuit [2].

Figure 2.6 typically illustrates how flow expands at a high divergence angle while internal devices such as screens or vanes break up large-scale turbulence structures. This helps maintain a more uniform velocity distribution and reduces the risk of boundary layer separation along the diffuser walls. The trade-off is increased pressure loss due to energy dissipation across the screens.

2.6.2 Wide-Angle Diffusers with Screens and Vanes

Where spatial constraints prevent the use of a long conventional diffuser, a wide-angle diffuser provides a compact alternative. A wide-angle diffuser uses a much larger divergence angle potentially exceeding 45° but incorporates internal flow conditioning elements to prevent separation. Two approaches are documented in the literature.

The first uses fine wire mesh screens stretched across the diffuser cross-section. The screens introduce a controlled pressure drop that stabilizes the flow and suppresses separation, though they increase overall head loss [3].

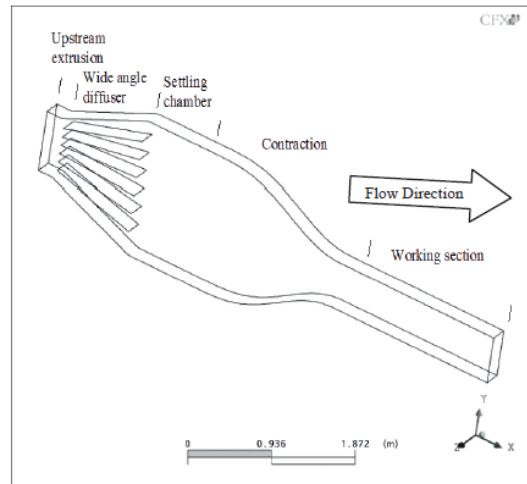


Figure 2.6: Internal guide vanes within a wide-angle diffuser

The second approach uses internal guide vanes that subdivide the diffuser passage into smaller channels, each with an effective diffusion angle at or below 7° . As illustrated in Figure 2.6, the guide vanes are positioned within the wide-angle diffuser to guide the flow through narrower passages, thereby reducing adverse pressure gradients and preventing boundary layer separation. Cochran and Kline demonstrated that six full-length flat vanes within a 49° wide-angle diffuser reduced the effective passage angle to 7° , achieving stable, attached flow throughout [9]. This approach significantly reduces the overall system length while maintaining good flow quality, making it a preferred solution for space-constrained hydrodynamic tunnel designs [3], [2].

2.7 Flow Conditioning - Honeycomb Structures and Screens

Flow conditioning components are placed upstream of the contraction section to reduce turbulence intensity, eliminate swirl, and promote a uniform velocity distribution before the flow enters the test section. In hydrodynamic tunnels, the two primary conditioning elements are honeycomb structures and wire mesh screens, housed within a settling chamber [3].

Honeycomb structures consist of arrays of parallel cells oriented along the flow direction, functioning to break down large-scale turbulent eddies into smaller, more rapidly decaying structures and to eliminate swirl generated by upstream bends and the pump. Hexagonal cells are preferred for their gapless tiling and uniform flow distribution [10]. The two governing design parameters are the turbulence reduction factor η and the pressure drop coefficient k , which conflict greater turbulence suppression increases pressure loss. The length-to-cell-

diameter ratio (L/D) is the primary design variable balancing these objectives, with values between 6 and 10 recommended for small/medium-scale applications [10], [8].

The settling chamber downstream of the honeycomb allows residual turbulence to decay before the contraction. A chamber length of 30 to 40 cell diameters is generally sufficient for this purpose [8], [3].

Wire mesh screens complement the honeycomb by targeting the axial component of turbulence, which honeycombs alone do not effectively suppress. Screens introduce a controlled pressure drop that redistributes momentum and breaks down large axial eddies. An open area ratio of at least 58 - 60% is recommended to avoid flow instability, and where multiple screens are used in series, a spacing of at least 50 to 75 mesh sizes between them allows the wake of each screen to decay sufficiently [3], [13].

Used in combination, honeycombs and screens produce a synergistic conditioning effect the honeycomb removes swirl and lateral fluctuations while the screens reduce axial turbulence together achieving turbulence intensities below 1% at the test section inlet in well-designed facilities [10], [3].

2.8 Head Loss Analysis Methods

Accurate estimation of head loss across each component of the tunnel circuit is necessary to determine the pump head requirement and ensure that the selected pump can sustain the desired flow velocity. Total head loss in a hydrodynamic tunnel circuit comprises both major losses, arising from skin friction along straight pipe and duct sections, and minor losses, arising from flow through individual components such as bends, diffusers, contractions, honeycombs, and screens [8].

Major losses are calculated using the Darcy-Weisbach equation:

$$h_f = f \left(\frac{L}{D_h} \right) \left(\frac{U^2}{2g} \right) \quad (3)$$

where f is the Darcy friction factor, L is the duct length, D_h is the hydraulic diameter, and U is the mean flow velocity. For turbulent flow, the friction factor is obtained from the Moody chart or the Colebrook-White equation as a function of Reynolds number and relative wall roughness [8].

Minor losses for each component are expressed in terms of a dimensionless loss coefficient K , such that,

$$h_m = K \frac{U^2}{2g} \quad (4)$$

Loss coefficients for standard components bends, contractions, diffusers, and screens are obtained from established empirical correlations and experimental data available in the literature. For components with complex geometry, such as honeycombs and wide-angle diffusers with vanes, loss coefficients are determined from component-specific design charts and validated CFD results [8], [3].

The total system head loss is the sum of all major and minor losses around the complete circuit. Comparing this total against the available pump head establishes whether the selected pump is sufficient and identifies which components contribute most significantly to energy loss. This analysis also informs component optimization decisions if total head loss is found to be small relative to available pump head, as was observed in several reviewed designs [3], it indicates that flow quality indicators such as turbulence intensity and velocity uniformity are more appropriate optimization criteria than head loss minimization alone.

2.9 Structural Considerations for Tunnel Fabrication

The structural design of hydrodynamic tunnel components must ensure that all parts can withstand the internal pressure exerted by the water column without excessive deformation or failure, while remaining practically manufacturable within the constraints of available facilities, budget, and material options [8].

The primary structural load on tunnel components is the hydrostatic pressure from the internal water column, which acts outward on all wetted surfaces. For a vertical closed-loop tunnel, this pressure varies with depth and must be evaluated at the worst-case location for each component. Wall thickness, material selection, and the use of reinforcing ribs or stiffeners are the principal means by which structural adequacy is achieved [3].

Material selection for small/medium-scale tunnel fabrication involves a trade-off between structural performance, water resistance, optical transparency, weight, and cost. Acrylic and tempered glass are commonly used for test section walls where optical access is required, offering good transparency and adequate strength for low-pressure applications [4]. For non-transparent structural components such as diffusers, settling chambers, and bends, materials

such as stainless steel, mild steel, and fiber-reinforced composites are documented in the literature. Glass fiber reinforced polymer (GFRP) in particular offers an attractive combination of high strength-to-weight ratio, corrosion resistance, and formability into complex shapes using relatively simple mould-based fabrication methods properties that make it well-suited to the fabrication of curved tunnel components in resource-constrained settings [8].

Finite Element Analysis (FEA) is widely used in tunnel component design to evaluate stress distributions and deformations under internal pressure loading, allowing wall thicknesses and reinforcement arrangements to be optimised before fabrication. Studies have shown that plain shell structures without reinforcement are often insufficient to meet deflection limits under realistic water column pressures, and that the addition of ribs or external stiffeners is typically necessary to achieve adequate structural performance within practical wall thickness constraints [3], [8].

2.10 Summary and Justification of Selected Design Direction

The literature review presented in the preceding sections establishes the theoretical and empirical basis upon which the design decisions for this project were made. Drawing from the reviewed configurations, design parameters, and fabrication approaches, each major design decision is summarized and justified in Table 2.2.

Table 2: Summary of selected design decisions and justifications

Design Decision	Selected Option	Justification
Tunnel orientation	Vertical	Minimizes floor space within 2.5 m height constraint; facilitates air bubble removal and gravity-assisted return flow
Circuit type	Closed-loop	Ensures stable, repeatable flow conditions with minimal water consumption
Test section type	Open-channel	Eliminates pressure sealing requirements; reduces fabrication complexity and cost; allows direct access for visualization and probes
Contraction ratio	$\sim 7:1$	Literature consensus for small/medium-scale low-turbulence tunnels; balances flow quality against settling chamber size
Contraction contour	Bell-Mehta 5th-degree polynomial	Ensures zero wall slope and curvature at both ends; prevents separation-induced turbulence
Diffuser type	Wide-angle with internal guide vanes	Satisfies spatial constraints while maintaining flow quality; reduces effective passage angle to 7° per channel
Flow conditioning	Honeycomb ($L/D = 6-10$) + wire mesh screens	Suppresses lateral and axial turbulence components achieves turbulence intensity below 1%
Fabrication material	Glass fiber reinforced polymer (GFRP)	High strength-to-weight ratio; corrosion resistant; formable into complex shapes within available budget

The pump procured by the department delivers a flow rate of $0.0136 \text{ m}^3/\text{s}$, which together with the selected test section dimensions produces a Reynolds number range of 0 to approximately 8,000 based on cylinder diameter sufficient to reproduce all principal external flow regimes around a circular cylinder from laminar steady flow through to transitional unsteady flow.

3 Chapter 3 - Methodology

3.1 Summary of Methodology

The methodology adopted in this project follows a structured, sequential design process comprising four main phases: initial design and analytical calculations, simulation-based optimization, detailed design, and fabrication. This approach ensures that each design decision is systematically justified before progressing to the next phase, minimizing the risk of costly design revisions during fabrication.

In the first phase, the tunnel configuration was selected through a comparative analysis of existing hydrodynamic tunnel designs as documented in the literature. Following configuration selection, analytical calculations were carried out for each major component to establish preliminary dimensions based on fluid dynamics principles and empirical design guidelines. A MATLAB-based head loss model was developed to evaluate the total system head loss and verify pump adequacy. Flow visualization using the hydrogen bubble technique was also identified and integrated into the design at this stage.

In the second phase, single-phase CFD simulations were carried out in ANSYS Fluent to evaluate and optimize the flow behavior within critical components, particularly the wide-angle diffuser and vertical diffusers. Finite Element Analysis (FEA) was conducted in parallel to assess the structural integrity of glass fiber components under internal water column pressure and to determine the reinforcement requirements.

In the third phase, the insights gained from analytical calculations and simulation results were consolidated into a detailed design, comprising finalized component dimensions, a complete CAD model, and engineering drawings suitable for fabrication.

In the fourth phase, all primary structural components were fabricated using glass fiber and steel bars and sheets, following the specifications established in the detailed design phase. Assembly of the complete tunnel circuit and experimental validation are carried out as the final stage of the project.

3.2 Initial Design and Analytical Calculations

The initial design phase comprised four sequential activities: configuration selection, component design and sizing, head loss analysis, and flow visualization implementation. Each activity informed the next, with the outputs of each stage feeding directly into the simulation and optimization phase described in chapter 4.

3.2.1 Configuration Selection Approach

The tunnel configuration was selected through a systematic review of existing hydrodynamic tunnel facilities documented in the literature, followed by evaluation against a defined set of project constraints. The primary constraints governing the selection were:

- Available pump flow rate: $Q = 0.0136 \text{ m}^3/\text{s}$ (13,000 gph)
- Total material budget: LKR 200,000
- Maximum vertical height: 2.5 m
- Fabrication limited to locally available materials and methods
- Target Reynolds number range: $Re = 0 - 60,000$ (based on cylinder diameter 20mm to 10 mm)

Four configuration categories were identified and reviewed: small/medium-scale open-channel horizontal, small/medium-scale closed-channel horizontal, small/medium-scale closed-channel vertical, and large-scale closed-channel horizontal. Each was evaluated against the project constraints using the criteria summarized in Table 3.1.

Table 3: Evaluation of tunnel configurations against project constraints

Criterion	Open-Channel Horizontal	Closed-Channel Horizontal	Closed-Channel Vertical	Large-Scale Horizontal
Max. achievable velocity	Low (~0.5 m/s)	Moderate	Moderate-High	High
Flow quality	Poor-Moderate	Good	Good	Excellent
Pressure control	No	Limited	Limited	Yes
Fabrication complexity	Low	Moderate	High	Very High
Space requirement	Large floor area	Large floor area	Small floor area	Very large
Budget compatibility	✓	✓	✓	✗
Height constraint (2.5 m)	✗	✗	✓	✗

The vertical closed-channel configuration with an open-channel test section was identified as the only option satisfying all constraints simultaneously. The Reynolds number feasibility of the selected configuration was confirmed using the continuity equation and the external flow Reynolds number formulation:

$$U_{ts} = \frac{Q}{A_{ts}} \quad (5)$$

$$Re = \frac{U_{ts} \cdot D}{\nu} \quad (6)$$

where A_{ts} is the test section cross-sectional area, D is the cylinder diameter, and $\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$ is the kinematic viscosity of water at 20°C. Two test section sizes were evaluated as shown in Table 3.2.

Table 4: Reynolds number feasibility check for candidate test section sizes

Test Section Size (m)	Area (m ²)	U _{ts} (m/s)	Re (D = 50 mm)
0.30 × 0.30	0.09	0.151	~7,550
0.15 × 0.15	0.023	0.604	~30,200

The 0.15 × 0.15 m test section was selected as it produces a Reynolds number of approximately 30,200 at full pump flow for a 50 mm diameter cylinder - sufficient to reproduce all target flow regimes from laminar steady flow ($Re < 5$) through to transitional unsteady flow ($Re < 3 \times 10^5$), as required by the experimental objectives [2], [3].

3.2.2 Component Design and Sizing

The tunnel circuit was divided into its primary functional components, with each member responsible for the analytical design of assigned components. The design of each component followed a sequential approach: establishing governing equations from literature, selecting key design parameters within recommended ranges, and computing final dimensions based on the available flow rate $Q = 0.0136 \text{ m}^3/\text{s}$. The following subsections present the analytical design methodology for the contraction section and interconnecting pipelines.

Wide-Angle Diffuser Design

In many established references, a divergence half-angle of 7° is recommended as the maximum angle for a diffuser to maintain attached flow and achieve efficient pressure recovery [2]. However, adopting a 7° divergence angle would result in an excessively long diffuser that is not practical within the physical constraints of the tunnel layout, and it is difficult to achieve a large pressure recovery within a compact length at this angle. Therefore, a wide-angle diffuser configuration was selected to achieve the required pressure recovery within a compact length.

The primary function of the wide-angle diffuser is to decelerate the flow and convert kinetic energy into static pressure before the flow enters the honeycomb flow straightener and settling chamber. The key design parameters of the wide-angle diffuser are summarized in Table 1, and were established with reference to the foundational work of Reneau, Johnston, and Kline [11] on the performance and design of straight, two-dimensional diffusers, and Waitman, Reneau, and Kline [12] on the effects of inlet conditions on diffuser performance. Given the large divergence angle adopted, flow separation was anticipated within the diffuser. CFD simulation was therefore carried out to assess the extent of flow separation and its effect on the downstream flow quality. If the simulation revealed that the separation significantly degraded the flow conditions, splitter vanes would be introduced into the diffuser as a corrective measure. The CFD analysis and its findings are discussed in detail in Chapter 4.

Table 5: Wide angle diffuser -Design parameters

Parameter	Description	Value
2θ	Total divergence angle	35°
AR	Area ratio	3
W_{in}	Inlet width	150 mm
W_{out}	Outlet width	450 mm
N	Diffuser length	475.74 mm
N/W	Length to inlet width ratio	3.17

Vertical Diffuser Design

The vertical transition section serves a dual purpose in the hydrodynamic tunnel circuit. Its primary function is to decay the turbulence generated by the pump before the flow reaches the downstream components. Additionally, it is specifically designed to counteract the gravitational pressure drop that occurs as the fluid ascends vertically. In a standard vertical pipe of constant diameter, static pressure decreases significantly as the fluid rises due to the hydrostatic head associated with the weight of the fluid column. By expanding the cross-sectional area along the flow direction, the transition section converts a portion of the fluid's dynamic pressure into static pressure, achieving pressure recovery that compensates for this gravitational loss. The finalized design parameters are summarized in Table 3.4.

Table 6: Design Parameters

Parameter	Value
Length (L)	613.12 mm
Inlet Diameter (W)	75 mm
Total Divergence Angle	7°
Area Ratio (AR)	5.09
Length-to-Width Ratio (L/W)	8.17 (< 25)
Outlet Dimensions	150 mm × 150 mm

The selection of the total divergence angle of 7° and the length-to-inlet-width ratio are both grounded in established diffuser design literature. The relationship between diffuser geometry and performance is primarily based on the works of Reneau et al. [11], Kline et al., and Sagi and Johnston. Their studies demonstrate that optimum diffuser effectiveness is achieved at a total divergence angle of approximately 7°, as this represents the threshold at which flow separation is minimized, thereby maximizing pressure recovery and overall diffuser performance. Furthermore, they established that the length-to-inlet-width ratio should not exceed 25 to ensure attached flow and acceptable performance. The designed L/W ratio of 8.17 comfortably satisfies this criterion. This selection is further supported by guidelines from [2], which independently arrived at the same optimal divergence angle of 7° for minimizing flow separation and maximizing pressure recovery.

Test section

The test section of the hydrodynamic tunnel is where the models are tested, and data are collected. One of the key objectives of our project is to achieve the desired flow conditions within the test section. Hydrodynamic tunnel test section can be divided various ways according to the research papers likes, open channel or closed channel, high speed or low speed and by their cross-section shape rectangular or circular. Most of the industrial level large scaled and medium level scaled high speed water tunnel used closed channel test section and some of the small scaled law speed water tunnel used open channel test section in their water tunnels. Based on literature studies, small laboratory-scale water tunnels commonly use rectangular open-channel test sections because they are easier to manufacture, provide good visualization capability, and reduce fabrication cost compared to closed pressurized sections.

The dimensions of the test section were selected according to the available pump capacity, desired Reynolds number range, blockage limitations, and laboratory space constraints. The test section was designed to operate mainly within the transitional unsteady flow region ($150 < Re < 3 \times 10^5$). Due to the limitation of the available university pump capacity, achieving Reynolds numbers higher than 3×10^5 is not practical. Increasing the Reynolds number beyond this range would require significantly higher flow velocities and pump power, resulting in increased system cost and pressure losses. Therefore, the test section dimensions and flow velocity were selected to achieve the maximum possible Reynolds number within the pump capability while still enabling meaningful hydrodynamic experiments and flow visualization.

Flow pattern	Reynolds number	Description
	$Re < 5$	No separation, laminar steady flow
	$5 < Re < 45$	Pair of vortices, laminar steady flow
	$45 < Re < 150$	Laminar vortex street, unsteady flow
	$150 < Re < 3 \cdot 10^5$	Transitional unsteady flow
	$3 \cdot 10^5 < Re < 3 \cdot 10^6$	Turbulent unsteady flow
	$Re > 3 \cdot 10^6$	Turbulent vortex street, unsteady flow

Figure 3.1: 2D cylinder flow at different Re numbers

The selected range is sufficient for studying important flow phenomena such as flow separation, vortex shedding, transition to turbulence, and wake formation around test models. Since the available pump can deliver a maximum flow rate of approximately $0.0136 \text{ m}^3/\text{s}$, the tunnel was designed as a small -scale low-speed water tunnel. A test section size of $0.15 \text{ m} \times 0.15 \text{ m}$ was selected to achieve desired Reynolds number range while maintaining manageable fabrication dimensions.

The average flow velocity inside the test section was determined using the continuity equation,

$$U = \frac{Q}{A} \quad (5)$$

Where:

- (U) = average flow velocity (m/s)
- (Q) = volumetric flow rate (m³/s)
- (A) = cross-sectional flow area (m²)

For the selected test section:

$$A = 0.15 \times 0.15 = 0.0225 \text{ m}^2$$

$$U = 0.0136 / 0.0225 \approx 0.604 \text{ m/s}$$

The Reynolds number was then calculated to evaluate whether the tunnel can achieve laminar-to-turbulent transitional flow conditions.

$$Re_D = \frac{UD}{\nu} \quad (7)$$

Where:

- (D) = characteristic length (hydraulic diameter of 50mm cylinder)
- (ν) = kinematic viscosity of water

Using (D = 0.15 m),

$$Re = (0.604 \times 0.05) / (1.0 \times 10^{-6}) \approx 30,200$$

This Reynolds number range is suitable for studying flow behavior in the transitional and turbulent regions for many hydrodynamic models.

Another important consideration in the test section design is the blockage ratio, which indicates how much of the flow area is occupied by the test model. Excessive blockage can accelerate the flow around the model and affect measurement accuracy. The blockage ratio is defined as,

$$\text{Blockage Ratio} = \frac{A_{\text{model}}}{A_{\text{test section}}} \quad (8)$$

For accurate experimental conditions, the blockage ratio should generally remain below 5–10% [20]. During the design stage, boundary layer growth within the test section was carefully analyzed. As water flows along the channel walls, viscous effects cause the boundary layer to thicken downstream, potentially reducing effective flow area and increasing velocity non-uniformity. Boundary layer growth was analyzed using MATLAB to minimize velocity non-uniformity. A 75 cm length was selected to allow flow stabilization, model placement, and measurement access while limiting pressure losses and boundary layer effects in the rectangular open-channel tunnel.

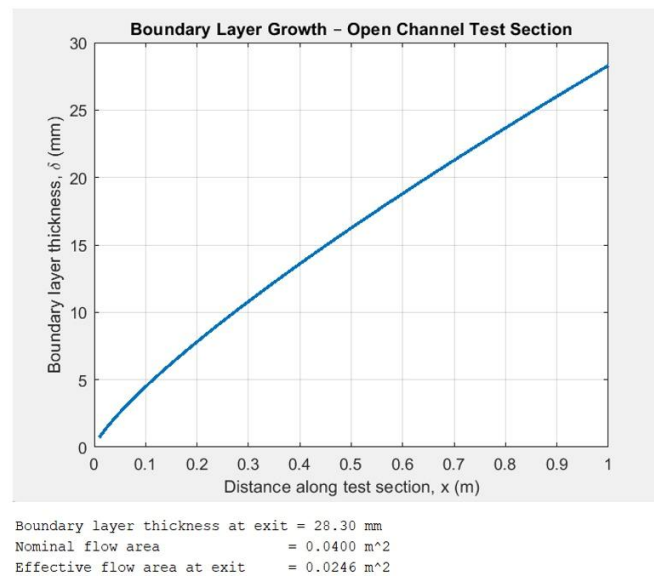


Figure 2: Boundary layer growth graph of open channel test section using MATLAB

Flow conditioning and honeycomb structure

To obtain accurate experimental results, the flow entering the test section must be uniform and stable. In water tunnels, honeycomb flow straightener was selected as the primary flow conditioning component because honeycombs are more suitable for water tunnels than screens. The honeycomb works by breaking large turbulent eddies into smaller, less energetic flow structures that decay quickly downstream, resulting in smoother and steadier flow before the contraction section. Its performance is evaluated using the turbulence reduction factor (η) and the pressure drop coefficient (k) [8].

Proper honeycomb cell sizing is essential to reduce turbulence while minimizing head loss. A settling chamber is placed downstream of the honeycomb to allow small-scale turbulence to decay before reaching the test section. Though preferred for uniform flow distribution,

though round cells may be used for cost reasons, potentially reducing uniformity. Adequate cell length ensures proper flow development and stabilization within each passage. For optimal flow uniformity, the honeycomb is positioned 30-40 cell diameters upstream of the contraction cone [8], [16].

Sizing Honeycomb Structure

The honeycomb cross section is sized to match the inlet face of the contraction cone.

Overall Width	600 mm
Overall Height	200 mm
Pump Flow Rate (Q)	13,000 gal/hr = 0.01367 m ³ /s
Average Flow Velocity (U)	$U = Q / A = 0.114 \text{ m/s}$

Cell Diameter Selection

The cell diameter (D) is selected so that, the honeycomb contains many cells across the duct, and the resulting cell Reynolds number exceeds the turbulent threshold ($Re_D > 2000$) [8].

The diameter Reynolds number is,

$$Re_D = \frac{UD}{\nu} \quad (9)$$

Most literature recommends cell diameters in the range of 5 - 20 mm using commercially available polymer or aluminum honeycomb. Hexagonal cells are preferred as they eliminate spatial gaps and ensure uniform flow distribution. Turbulent cell flow ($Re_D > 2000$) is preferred over laminar because it produces a more uniform exit velocity profile reducing velocity and pressure gradients downstream and therefore less residual turbulence in the test section.

Selected Cell Diameter (D)	10 mm
Diameter Reynolds Number (Re_D)	$Re_D = (0.114 \times 0.010) / 1.004 \times 10^{-6} \approx 2200$
Turbulence Criterion	$Re_D = 2200 > 2000$
Cell Shape	hexagonal preferred

The Reynolds number based on diameter and length, together with the length to diameter ratio, can be used to determine the pressure drop coefficient using Figure 3.3 (Chart 2). Additionally, the ratio of initial integral length scale to cell length, along with the pressure drop coefficient, can be used to estimate the turbulence intensity using Figure 3.3 (Chart 1) [8].

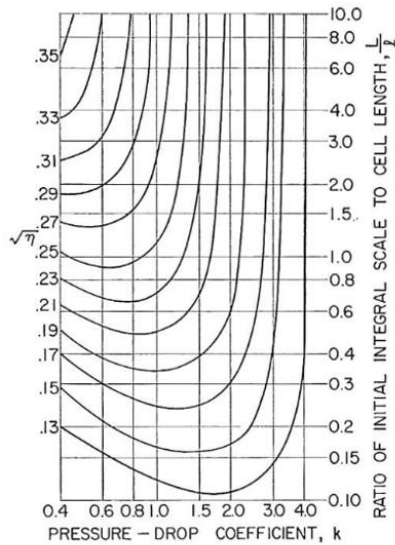


Fig. 1 Isoreduction contours as a function of pressure-drop coefficient and ratio of integral scale of initial turbulence to cell length

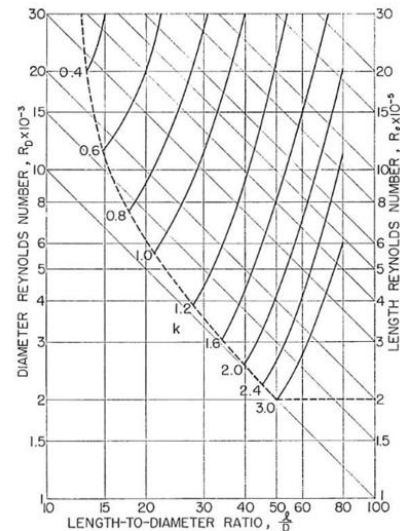


Fig. 2 Pressure-drop coefficient and length Reynolds number as a function of diameter Reynolds number and length-to-diameter ratio. Flow is fully developed above and to the right of the dashed line.

Figure 3.3: J.L. Lumley, J.F. McMahon - Reducing Water Tunnel Turbulence by Means of a Honeycomb [8]

Honeycomb Length Selection

The honeycomb length (l) is governed by the length-to-diameter ratio l/D . Practical designs require $l/D \geq 5$ to ensure effective turbulence suppression. A length of $l = 500$ mm is selected to achieve $l/D = 50$, placing the operating point well within the fully developed turbulent flow regime on the Lumley & McMahon chart.

Selected Honeycomb Length (l)	500 mm
Length-to-Diameter Ratio (l/D)	$500 / 10 = 50$
Length Reynolds Number (Re_l)	$Re_l = (0.114 \times 0.50) / 1.004 \times 10^{-6} \approx 10^5$
Flow Regime	Fully developed turbulent cell flow

Pressure-Drop Coefficient

Using $Re_D \approx 2200$ and $L/D = 50$ as coordinates, the pressure-drop coefficient k is read from the Lumley & McMahon Fig. 2 chart (Pressure-drop coefficient vs. length-to-diameter ratio). The operating point falls above and to the right of the dashed line, confirming fully developed flow within the cells.

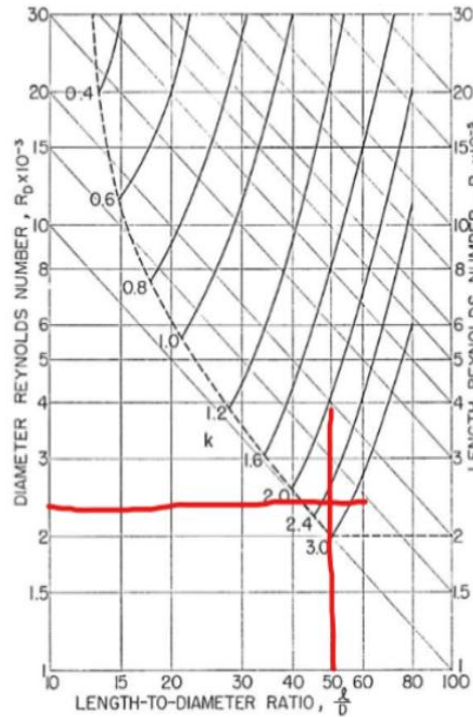


Figure 4: Variation of pressure-drop coefficient and length Reynolds number as a function of diameter Reynolds number and length-to-diameter ratio, indicating fully developed flow region above and to the right of the dashed line. The pressure loss across the honeycomb [8]

$$\Delta p = \frac{k \rho U^2}{2} \quad (10)$$

Pressure-Drop Coefficient (k)	2.4 (from Lumley & McMahon)
Water Density (ρ)	1000 kg/m ³
Dynamic Pressure ($\frac{1}{2}\rho U^2$)	$\frac{1}{2} \times 1000 \times 0.114^2 = 6.50$ Pa
Pressure Loss (Δp)	$\Delta p = 2.4 \times 6.50 \approx 15.6$ Pa

Integral Turbulence Length Scale (L)

The integral turbulence length scale L is a measure of the largest turbulent structures in the approaching flow. As a conservative engineering estimate for the current facility, L is taken as half the hydraulic diameter at the honeycomb inlet consistent with the approach used by

Libin Daniel (Embry-Riddle Aeronautical University) and supported by Lumley & McMahon ($L \approx 0.3\text{--}0.5 \times$ upstream duct diameter).

The hydraulic diameter of the rectangular honeycomb section is:

$$D_h = \frac{4A}{P} = 0.30 \text{ m} \quad (11)$$

$$L = \frac{D_h}{2} = 0.15 \text{ m} \quad (12)$$

Honeycomb Area (A)	$0.60 \times 0.20 = 0.12 \text{ m}^2$
Wetted Perimeter (P)	$2 \times (0.60 + 0.20) = 1.60 \text{ m}$
Hydraulic Diameter (D_h)	0.30 m
Integral Length Scale (L)	$L = 0.15 \text{ m}$
L / l Ratio	$0.15 / 0.50 = 0.30$

Turbulence Reduction Factor (η)

The turbulence reduction factor η is defined as the ratio of downstream to upstream mean-square velocity fluctuations.

$$\eta = \frac{u_{\text{downstream}}'^2}{u_{\text{upstream}}'^2} \quad (13)$$

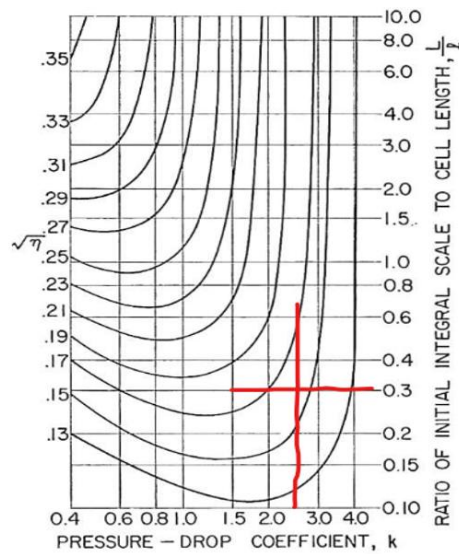


Figure 5 : Isoreduction contours showing the relationship between pressure-drop coefficient and the ratio of the integral length scale of initial turbulence to the cell length [8]

Entering the Lumley & McMahon Fig. 1 chart with the pressure-drop coefficient $k = 2.4$ and the L/l ratio = 0.30, the velocity-based turbulence reduction factor $\sqrt{\eta}$ is read off the y-axis.

$$\sqrt{\eta} \approx 0.15 \rightarrow \eta \approx 0.023$$

A value of $\sqrt{\eta} = 0.15$ means the RMS turbulent velocity downstream of the honeycomb is only 15% of the upstream value – an 85% reduction in turbulence intensity. This confirms the honeycomb design is highly effective [8].

k (pressure-drop coefficient)	2.4
L/l ratio	0.30
Chart Reference	Lumley & McMahon Fig. 1
$\sqrt{\eta}$ (velocity reduction factor)	≈ 0.15
η (energy reduction factor)	≈ 0.023 (~97.7% energy reduction)
Turbulence Intensity Reduction	~85% reduction in RMS turbulent velocity

Based on Lumley and McMahon design charts, the selected honeycomb geometry effectively balances pressure loss and turbulence reduction. It is expected to suppress turbulence intensity and reduce upstream swirl, producing uniform and stable flow. Overall, the flow conditioning system improves flow quality at the test section, ensuring reliable and repeatable hydrodynamic measurements.

Settling chamber design

In hydrodynamic tunnel design, honeycombs break down turbulent eddies and remove swirl, but each cell generates small wakes and shear layers that leave the flow slightly turbulent. The settling chamber, positioned downstream of the honeycomb, allows these residual disturbances to decay naturally before reaching the contraction cone and test section. Without adequate settling length, residual eddies entering the contraction are amplified by accelerating flow, causing flow separation, increased turbulence intensity, and compromised experimental results. The settling chamber is therefore essential for achieving uniform, low-turbulence flow in the test section.

Design Criteria and Sizing

The primary design parameter governing the settling chamber is its length, which is directly linked to the honeycomb cell diameter. Honeycomb specifications require that the honeycomb must terminate 30 to 40 cell diameters upstream of the contraction cone

entrance. For the current design, a honeycomb cell diameter (D) is 10 mm. Applying the 30 to 40 cell diameter rule gives,

- Minimum settling length: $30 \times 10 \text{ mm} = 300 \text{ mm}$

The settling chamber maintains a constant rectangular cross-section of $45 \text{ cm} \times 15 \text{ cm}$, matching the honeycomb outlet and contraction inlet to prevent sudden area changes. As flow exits the honeycomb, three simultaneous processes occur within the chamber: turbulence decays through viscous dissipation, the velocity profile becomes uniform, and residual swirl attenuates. Turbulent cell flow is preferred as it produces a flatter velocity profile, accelerating the decay process. Together with the honeycomb, the settling chamber forms the flow conditioning assembly, ensuring reliable, low-turbulence flow reaches the test section for accurate experimental measurements.

Contraction Section

The contraction section accelerates flow from the settling chamber cross-section to the test section, while suppressing upstream turbulence through streamwise flow acceleration. The design followed the methodology established by Bell and Mehta [1] and validated in published water tunnel designs [3], [2].

Contraction Geometry

The contraction inlet dimensions were determined by the upstream settling chamber geometry: width $W_{in} = 0.45 \text{ m}$, height $H_{in} = 0.15 \text{ m}$, giving an inlet area:

$$A_{in} = W_{in} \times H_{in} = 0.45 \times 0.15 = 0.0675 \text{ m}^2 \quad (14)$$

The test section exit dimensions are $0.15 \times 0.15 \text{ m}$, giving:

$$A_{ts} = 0.15 \times 0.15 = 0.0225 \text{ m}^2$$

The resulting contraction ratio:

$$r_c = \frac{A_{in}}{A_{ts}} = \frac{0.0675}{0.0225} = 3 \quad (15)$$

This is a two-dimensional contraction — the width reduces from 0.45 m to 0.15 m while the height remains constant at 0.15 m. A contraction ratio of 3:1, while lower than the 6–10 range typically recommended for wind tunnels, is consistent with published water tunnel designs where heavier construction requirements and spatial constraints limit the achievable ratio [3]. Sargison et al. employed an identical contraction ratio of 3:1 in a recirculating water

tunnel designed for boundary layer research, demonstrating acceptable flow uniformity at the test section inlet [3].

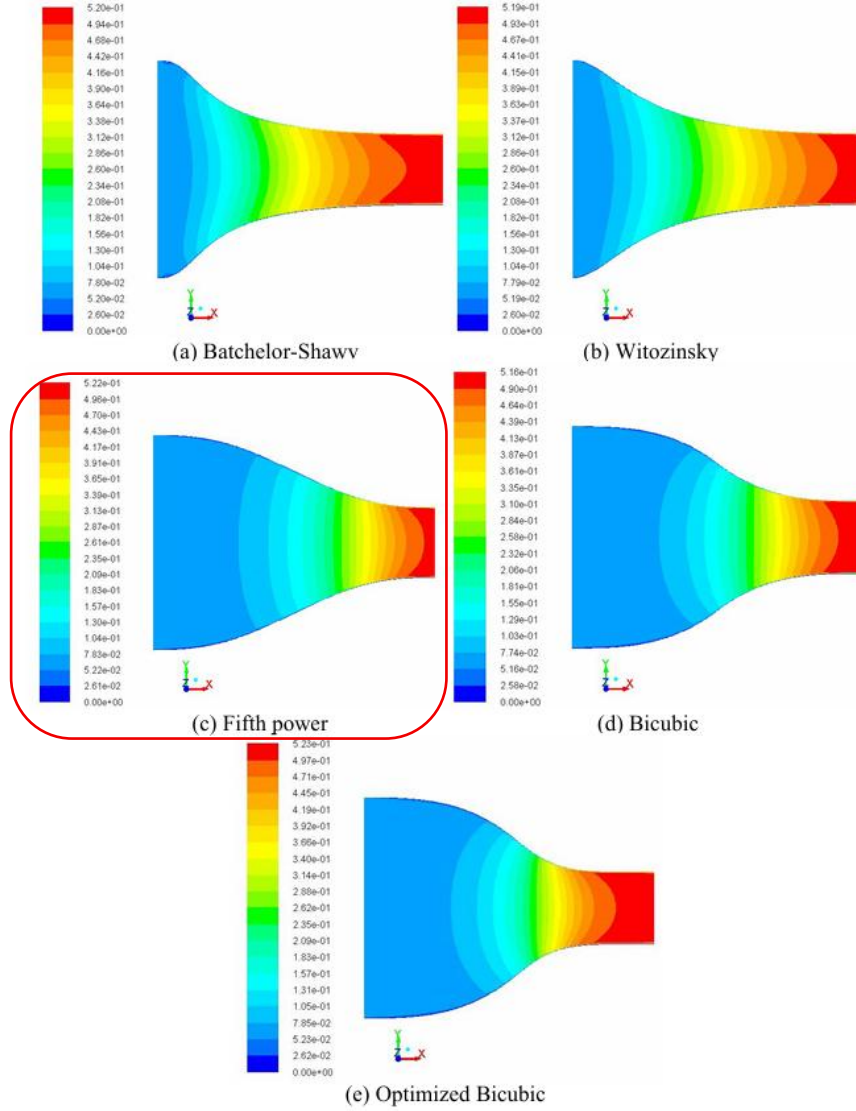


Figure 3.6: Bell-Mehta fifth-degree polynomial wall profile for the contraction section
Source: Y. Joang et al., 2018 [18]

Flow Velocities

From continuity:

$$U_{ts} = \frac{Q}{A_{ts}} = \frac{0.0136}{0.0225} = 0.604 \text{ m/s} \quad (16)$$

$$U_{in} = \frac{Q}{A_{in}} = \frac{0.0136}{0.0675} = 0.201 \text{ m/s} \quad (17)$$

The test section Reynolds number based on hydraulic diameter:

$$Re_{ts} = \frac{U_{ts} \cdot D_h}{\nu} = \frac{0.604 \times 0.15}{1.0 \times 10^{-6}} \approx 90,600 \quad (18)$$

For external flow over a cylinder of diameter $D = 13$ mm at maximum tunnel velocity:

$$Re_{cyl} = \frac{U_{ts} \cdot D}{\nu} = \frac{0.604 \times 0.013}{1.0 \times 10^{-6}} \approx 7,850 \quad (19)$$

confirming the target Reynolds number range of 0–8,000 is achievable with the selected test section dimensions.

Contraction Length

The contraction length l was determined using the width-to-length ratio guidance of Bell and Mehta [1]:

$$0.667 \leq \frac{W_{in}}{l} \leq 1.79 \quad (20)$$

Taking $W_{in} = 0.45$ m and a width-to-length ratio of 1.0 for a compact geometry:

$$l = \frac{W_{in}}{1.0} = 0.45 \text{ m} \quad (21)$$

Contraction Wall Profile

Since the contraction operates only in the width direction, the Bell-Mehta fifth-degree polynomial was applied to the width profile [1]:

$$y(x) = H_i - (H_i - H_e) \left[6 \left(\frac{x}{l} \right)^5 - 15 \left(\frac{x}{l} \right)^4 + 10 \left(\frac{x}{l} \right)^3 \right] \quad (22)$$

where:

- $y(x)$ = local half-width at axial position x from inlet (m)
- H_i = inlet half-width = $0.45/2 = \mathbf{0.225 \text{ m}}$
- H_e = exit half-width = $0.15/2 = \mathbf{0.075 \text{ m}}$
- l = contraction length = $\mathbf{0.45 \text{ m}}$

This profile ensures zero wall slope and zero curvature at both inlet and exit, eliminating inflection-point-induced boundary layer separation [1], [2].

Pressure Drop Check

Applying Bernoulli's equation between contraction inlet (1) and test section exit (2), neglecting minor losses for a smooth contraction:

$$\Delta p = \frac{1}{2} \rho (U_{ts}^2 - U_{in}^2) = \frac{1}{2} \times 1000 \times (0.604^2 - 0.201^2) \approx 162 \text{ Pa} \quad (23)$$

This pressure drop is small relative to the available pump head and does not represent a significant energy demand on the system.

The finalized contraction section parameters are summarized in Table 3.5.

Table 7: Finalized contraction section design parameters

Parameter	Value
Inlet dimensions	0.45 m (W) $\times$ 0.15 m (H)
Exit dimensions	0.15 m $\times$ 0.15 m
Inlet area A_{in}	0.0675 m ²
Exit area A_{ts}	0.0225 m ²
Contraction ratio r_c	3
Contraction length l	0.45 m
Wall profile	Bell-Mehta 5th-degree polynomial
Inlet half-width H_i	0.225 m
Exit half-width H_e	0.075 m
Inlet velocity U_{in}	0.201 m/s
Exit velocity U_{ts}	0.604 m/s
Static pressure drop Δp	$\sim$ 162 Pa

Interconnecting Pipeline Selection

The interconnecting pipelines link the primary structural components of the tunnel circuit. Their design was governed by three criteria: maintaining acceptable flow velocity, minimizing head loss, and compatibility with commercially available PVC pipe sizes.

The mean flow velocity in a circular pipe of diameter d is:

$$U_{pipe} = \frac{4Q}{\pi d^2} \quad (24)$$

The Darcy-Weisbach equation gives head loss per unit length:

$$\frac{h_f}{L} = \frac{f \cdot U_{pipe}^2}{2gd} \quad (25)$$

where f is the Darcy friction factor from the Colebrook-White equation. Three standard PVC pipe diameters were evaluated as shown in Table 3.6.

Table 8: Head loss evaluation for candidate pipe diameters

Diameter (mm)	U_{pipe} (m/s)	Re	$h_f /$ L (m/m)	Verdict
50	6.92	346,000	1.18	✗ Excessive velocity and loss
75	3.07	230,000	0.144	✓ Selected
100	1.73	173,000	0.033	✗ Unnecessary bulk and cost

A diameter of **75 mm** standard uPVC pipe was selected, providing moderate flow velocity, manageable head loss, and availability through local suppliers. The Reynolds number in the pipe:

$$Re = \frac{U_{pipe} \cdot d}{\nu} = \frac{3.07 \times 0.075}{1.0 \times 10^{-6}} \approx 2.3 \times 10^5 \quad (26)$$

confirms fully turbulent flow, consistent with the Darcy-Weisbach friction factor assumption [8].

Table 9: Selected pipeline specifications

Parameter	Value
Selected diameter	75 mm
Material	Standard uPVC
Flow velocity	3.07 m/s
Reynolds number	$\sim 2.3 \times 10^5$
Head loss per meter	0.144 m/m

3.2.3 Head Loss Analysis

A systematic head loss analysis was carried out to determine the total hydraulic resistance of the tunnel circuit and verify that the available pump head is sufficient to sustain the design flow rate of $Q = 0.0136 \text{ m}^3/\text{s}$. The analysis was first performed manually using established fluid mechanics equations, then implemented in a MATLAB model to enable rapid parametric evaluation during the component design iterations.

Head Loss Framework

The total head loss around the circuit H_{total} is the sum of major losses h_f due to friction in straight pipe sections, and minor losses h_m due to flow through individual components:

$$H_{total} = \sum h_f + \sum h_m \quad (27)$$

Major Losses

Major losses in straight pipe and duct sections were calculated using the Darcy-Weisbach equation:

$$h_f = f \cdot \frac{L}{D_h} \cdot \frac{U^2}{2g} \quad (28)$$

where f is the Darcy friction factor, L is the section length, D_h is the hydraulic diameter, and U is the mean flow velocity. For turbulent flow in smooth-walled PVC pipes, f was obtained from the explicit Swamee-Jain approximation to the Colebrook-White equation [8]:

$$f = \frac{0.25}{\left[\log_{10} \left(\frac{\varepsilon}{3.7D_h} + \frac{5.74}{Re^{0.9}} \right) \right]^2} \quad (29)$$

where ε is the pipe roughness height (taken as $\varepsilon = 0.0015$ mm for smooth uPVC).

Minor Losses

Minor losses for each tunnel component were expressed using the dimensionless loss coefficient K :

$$h_m = K \cdot \frac{U^2}{2g} \quad (30)$$

where U is the mean flow velocity at the component inlet. Loss coefficients were obtained from established empirical correlations and component-specific design guidelines as summarized in Table 3.8.

Table 10: Loss coefficients for tunnel circuit components

Component	Loss Coefficient (K)	Reference
90° cascade bend with vanes	0.10 – 0.20	[2]
Wide-angle diffuser (49°, 6 vanes)	0.30 – 0.40	[3]
Vertical diffuser (7° half-angle)	0.15 – 0.25	[1]
Contraction section	0.04	[4]
Honeycomb structure	0.40 – 0.80	[5]
Wire mesh screen	0.20 – 0.40	[5]
Square-round transition	0.05	[1]
PVC pipe sections	Darcy-Weisbach	[1]

Total System Head Loss

Applying the loss framework around the complete tunnel circuit, the total head loss was expressed as:

$$H_{total} = \sum_{i=1}^n K_i \cdot \frac{U_i^2}{2g} + \sum_{j=1}^m f_j \cdot \frac{L_j}{D_{h,j}} \cdot \frac{U_j^2}{2g} \quad (31)$$

where subscripts i and j denote individual minor loss components and pipe sections respectively. All velocities U_i were computed from continuity at each cross-section using $Q = 0.0136 \text{ m}^3/\text{s}$.

Pump Adequacy Check

The required pump head H_{pump} must satisfy:

$$H_{pump} \geq H_{total} + \Delta z \quad (32)$$

where Δz is the static head difference between the pump inlet and outlet due to the vertical tunnel geometry. The available pump head was obtained from the pump characteristic curve supplied with the procured pump.

MATLAB Model

A MATLAB-based head loss model was developed to automate the calculation across all components and enable sensitivity analysis during design iteration. The model accepts component dimensions as inputs, computes the flow velocity at each cross-section from continuity, evaluates major and minor losses for each element, and outputs the total circuit

head loss alongside a component-by-component breakdown. This allowed the influence of individual design parameters — such as pipe diameter, diffuser angle, and honeycomb dimensions — to be assessed rapidly without manual recalculation.

The structure of the MATLAB model is illustrated in Figure below, and the full model code is provided in Appendix B.

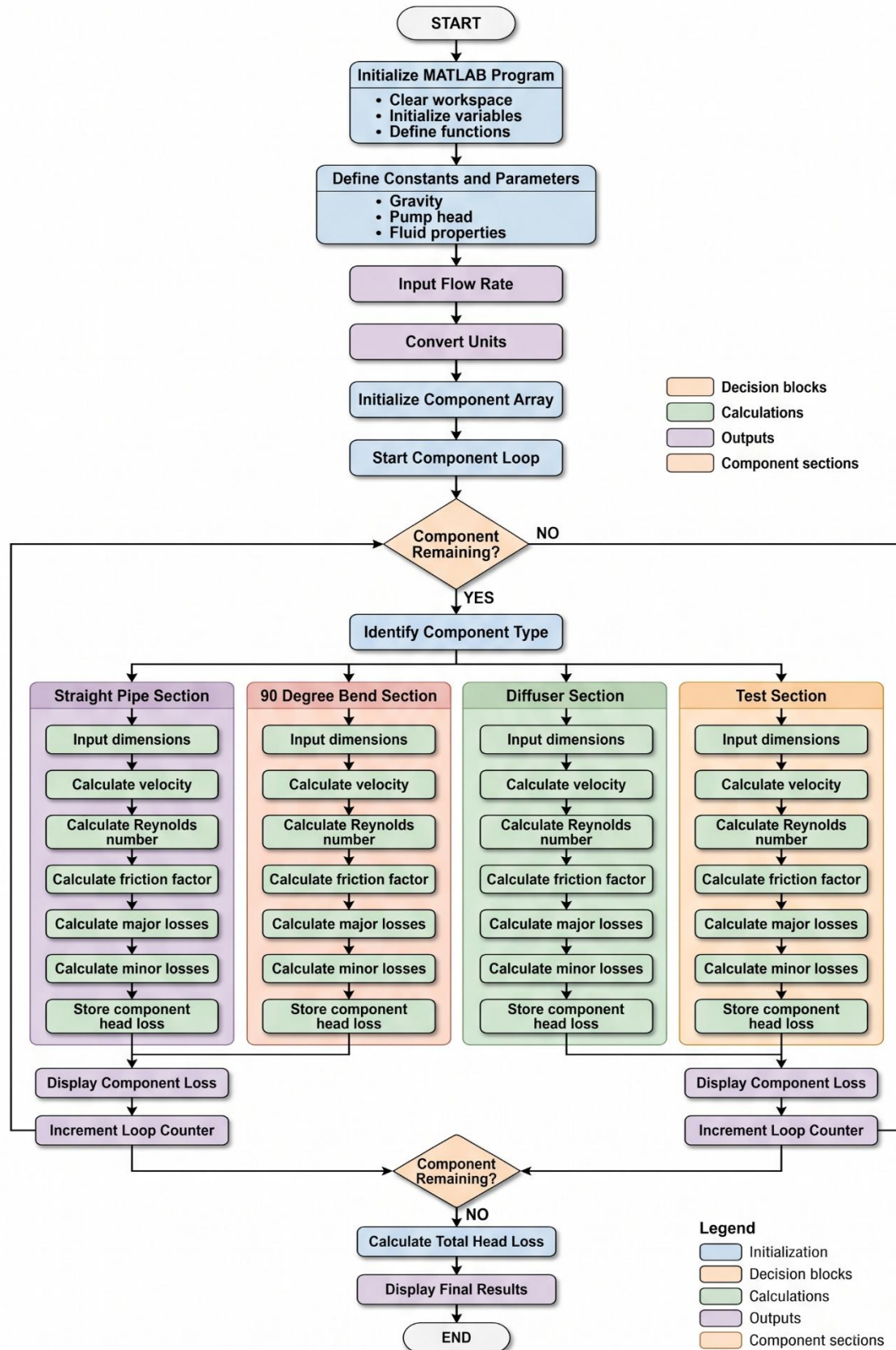


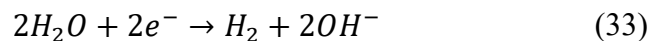
Figure 3.7: Flowchart of the MATLAB head loss model

3.2.4 Flow Visualization Implementation

Flow visualization using the hydrogen bubble technique was selected and integrated into the test section design during the initial design phase. The selection of this method over the more commonly used dye injection approach was driven by a fundamental constraint of closed-loop tunnel operation: dye injected into the circuit accumulates with repeated experimental runs, progressively contaminating the water and degrading flow quality. The hydrogen bubble method avoids this problem entirely, as the bubbles are generated in-situ and dissolve downstream without leaving any residual contamination [17].

Operating Principle

Hydrogen bubbles are generated by electrolysis using a thin stainless steel wire stretched across the flow direction in the test section. The wire acts as the cathode (negative electrode) in a DC electrical circuit, with the tunnel structure or a separate electrode serving as the anode. When current is applied, water molecules at the wire surface dissociate according to:



producing a continuous stream of small hydrogen bubbles that are carried downstream by the flow, making the velocity field directly visible [[17].

Wire Specifications and Placement

The wire diameter is a critical parameter — finer wires produce smaller, more uniform bubbles that follow the flow more faithfully. A wire diameter in the range of 25–50 μm is recommended for low Reynolds number applications [17]. The wire was positioned transversely across the full width of the test section, upstream of the cylinder model location, to generate a uniform bubble sheet that reveals the approaching flow structure and its subsequent development around the model.

Electrical Circuit

The circuit consists of a DC power supply, the cathode wire, and a return electrode. Applied voltage and current are adjusted to control bubble generation rate and density. Typical operating parameters for water tunnel hydrogen bubble systems are in the range of 10–50 V DC, with current adjusted to produce a visible but non-disruptive bubble density [17].

Imaging Setup

Adequate backlighting or side-lighting is required to make the bubble sheet visible against the test section background. A high-contrast background and LED lighting directed perpendicular to the bubble sheet provides sufficient visibility for qualitative flow pattern observation and photographic recording.

The integration of the hydrogen bubble system into the test section is illustrated in Figure 3.4.

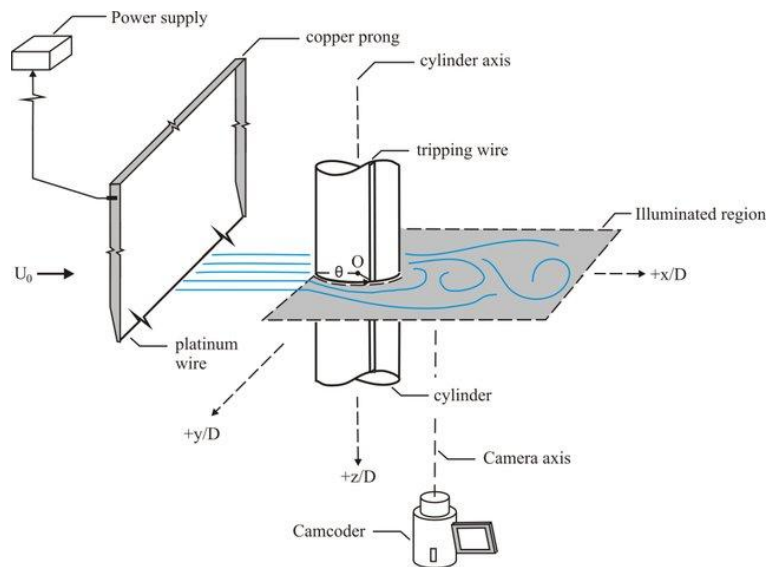


Figure 3.8: A representative schematic of the experimental setup employing the hydrogen bubble flow visualization system. The cylinder/wire and the platinum wire holder

4 Chapter 4 – Simulation, Optimization and detailed design

4.1 CFD Analysis Approach

4.1.1 Overview of Simulation Methodology

Role of Simulation in the Design Process

The hydrodynamic tunnel was designed primarily on the basis of established theoretical principles, empirical correlations, and guidance drawn from relevant literature. While this approach provided a well-founded starting geometry, it inherently involves simplifications that may not fully capture the complex three-dimensional flow behavior within the tunnel. CFD simulation was therefore employed as a complementary tool to validate the design and investigate specific flow phenomena that analytical methods alone could not adequately resolve.

The primary objective of the CFD analysis was to assess the internal flow characteristics of the critical components, specifically the wide-angle diffuser, the 90° bend, the settling chamber, the contraction section, and the test section. A key motivation was the known susceptibility of wide-angle diffusers to flow separation, which leads to reduced pressure recovery efficiency, increased turbulence intensity, and non-uniform velocity distributions downstream. The CFD analysis was therefore used to confirm the presence and extent of this separation and to evaluate design modifications, particularly the introduction of splitter vanes within the diffuser.

In summary, simulation served three distinct roles: as a validation tool to verify component performance, as a diagnostic tool to identify flow deficiencies, and as an optimization platform to guide design improvements prior to physical construction.

Selection of CFD Approach and Software

ANSYS Fluent was selected as the CFD solver for this study due to its widespread use in engineering applications and its well-validated turbulence modelling capabilities. The SST $k-\omega$ turbulence model was adopted for all simulations, as it is well suited for flows involving adverse pressure gradients, boundary layer separation, and recirculation zones, all of which are characteristic of flow through a wide-angle diffuser. Water was treated as an incompressible, Newtonian fluid with constant properties at ambient temperature, which is appropriate as the working fluid remains entirely in the liquid phase during normal tunnel operation.

Scope and Limitations of the Simulation

The simulation domain encompasses the key flow-conditioning components of the tunnel: the vertical diffuser, 90° bend, wide-angle diffuser, settling chamber, contraction section, and test section. Two notable simplifications were made relative to the physical tunnel. First, the recirculation pump was excluded from the domain, with a uniform velocity inlet boundary condition applied instead to represent the flow delivered by the pump. Second, the honeycomb flow straightener was not included, as resolving its individual cells would result in an excessively large mesh and prohibitive computational cost. Despite these simplifications, the simulation domain and boundary conditions are considered sufficiently representative to fulfil the primary objectives of validation and diffuser optimization.

4.1.2 Computational Fluid Dynamics (CFD) Methodology - Governing Equations and Turbulence Modelling

Governing Equations

The fluid flow through the hydrodynamic tunnel components was modelled by solving the Reynolds-Averaged Navier-Stokes (RANS) equations, which govern the conservation of mass and momentum for incompressible, viscous flows.

Continuity Equation (Conservation of Mass):

$$\frac{\partial U_i}{\partial x_i} = 0 \quad (34)$$

where U_i represents the time-averaged velocity component in the i -th coordinate direction, and x_i denotes the corresponding spatial coordinate. This enforces the incompressibility condition, valid for liquid water at the velocities encountered in this study.

RANS Equation:

$$\rho \left(U_j \frac{\partial U_i}{\partial x_j} \right) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \rho \overline{u'_i u'_j} \right] \quad (35)$$

where ρ is the fluid density, P is the mean static pressure, μ is the dynamic viscosity, and $-\rho u_i' \bar{u}_j'$ is the Reynolds stress tensor, accounting for the effect of turbulent velocity fluctuations on the mean flow.

Fluid Properties:

Water was treated as incompressible and Newtonian with constant properties at $\sim 20^\circ\text{C}$:

- $\rho = 998.2 \text{ kg/m}^3$
- $\mu = 1.003 \times 10^{-3} \text{ Pas}$.

The RANS approach was selected over LES or DNS, providing an appropriate balance between computational cost and accuracy for the engineering flow configurations in this study.

Turbulence Modelling - SST $k - \omega$

The SST $k-\omega$ (Shear Stress Transport) turbulence model, originally developed by Menter [14], was selected for all simulations in this study. The model solves two additional transport equations — one for the turbulent kinetic energy k , and one for the specific dissipation rate ω — which together provide the turbulent (eddy) viscosity μ_t required to close the RANS equations.

Transport Equations:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho U_j k)}{\partial x_j} = P_k - \beta^* \rho \omega k + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_k \mu_t) \frac{\partial k}{\partial x_j} \right] \quad (36)$$

$$\frac{\partial(\rho \omega)}{\partial t} + \frac{\partial(\rho U_j \omega)}{\partial x_j} = \frac{\gamma \rho}{\mu_t} P_k - \beta \rho \omega^2 + \frac{\partial}{\partial x_j} \left[(\mu + \sigma_\omega \mu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \frac{\rho \sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \quad (37)$$

Why SST k- ω was selected:

The SST k- ω model was chosen over alternative RANS models such as the standard k- ϵ or realizable k- ϵ for two principal reasons. First, the standard k- ϵ model is known to perform poorly in flows with strong adverse pressure gradients and boundary layer separation, which are the dominant flow features in the wide-angle diffuser. Second, the SST formulation improves upon the standard k- ω model by eliminating its sensitivity to free-stream turbulence boundary conditions, achieved through the blending function F_1 . The SST k- ω model is therefore well suited to the mixed flow regime present in this study attached boundary layers in the contraction and test section, and separated, recirculating flow in the diffuser.

4.1.3 Transient Simulation Approach

A transient simulation was performed for the wide-angle diffuser without vanes to capture the unsteady flow behavior within the diffuser. The transient approach was adopted as flow separation in a wide-angle diffuser is inherently unsteady in nature, the separated shear layers and recirculation zones fluctuate over time, and a steady-state RANS solution would not adequately represent this behavior. The transient simulation therefore resolves the time-dependent evolution of the separated flow, providing a physically representative picture of the flow field within the wide-angle diffuser.

Transient Simulation Settings:

The transient simulation used the PISO (Pressure-Implicit with Splitting of Operators) pressure-velocity coupling scheme, which is well suited to time-dependent flow problems due to its predictor-corrector structure and efficient handling of transient terms. The simulation was conducted in two phases of time stepping to balance accuracy and computational efficiency. In the initial phase, a fixed time step of $\Delta t = 0.001$ s was employed for the first 4 seconds of physical time to accurately capture the early transient development of the flow. Subsequently, due to computational resource limitations, the time

step was increased to $\Delta t = 0.01$ s for the remaining simulation duration, with a maximum of 40 iterations per time step to ensure sufficient convergence within each time increment.

The Courant-Friedrichs-Lewy (CFL) condition was used to assess the temporal resolution of the simulation, defined as:

$$CFL = \frac{V \times \Delta t}{\Delta x} \quad (38)$$

where V is the local flow velocity (m/s), Δt is the time step size (s), and Δx is the local cell size (m), approximated here as the cube root of the cell volume:

$$\Delta x = V_{cell}^{1/3} \quad (39)$$

Based on the maximum flow velocity of 3 m/s and minimum cell size of 4×10^{-4} m, the maximum CFL number was approximately **7.5** during the initial phase ($\Delta t = 0.001$ s) and approximately **75** during the second phase ($\Delta t = 0.01$ s), both within the acceptable range for the implicit PISO scheme. The simulation was run for a total physical time of 15 seconds, with convergence within each time step assessed using a normalized residual criterion of 10^{-3} for all solved variables, including velocity components, pressure, and turbulence quantities. Data sampling for time-averaged statistics was enabled throughout the simulation, with a total sampled time of approximately 15 s.

4.1.4 Simulation Domain Definition and Boundary Conditions

Computational Domain and Geometry

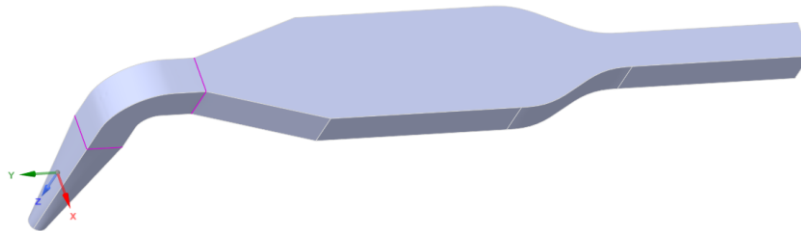


Figure 4.1 - Fluid domain

The computational domain includes the vertical transition diffuser, the 90° bend wide-angle diffuser, the settling chamber, the contraction section, and the test section. The honeycomb flow straightener was excluded from the domain as discussed in later sections.

Boundary Condition Setup

The boundary conditions were defined consistently across both simulations at the inlet, with differences applied at the outlet to suit the objectives of each simulation approach.

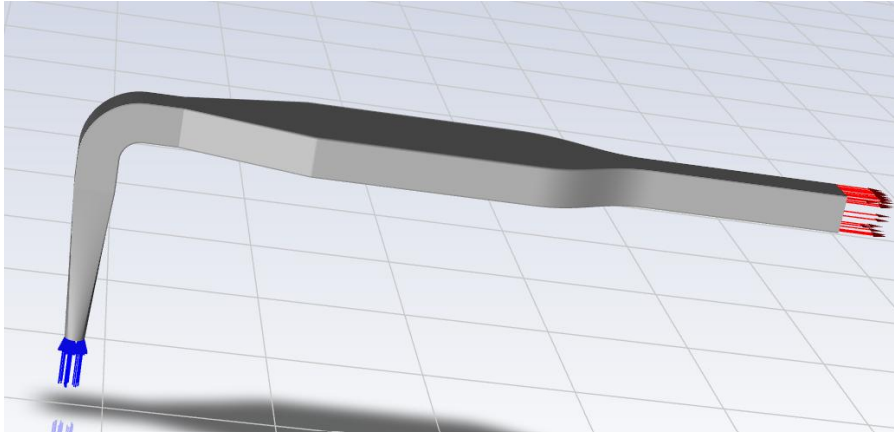


Figure 4.2 - Boundary condition setup

Inlet: A velocity inlet boundary condition was applied at the vertical transition diffuser inlet for both simulations, with a uniform inlet velocity of 3 m/s and turbulence intensity of 8%.

Outlet : A pressure outlet boundary condition was applied at the test section outlet for the transient simulation, with a gauge pressure of 0 Pa specified at the outlet plane. This condition is more appropriate for time-dependent simulations, allowing pressure fluctuations associated with unsteady flow separation to be naturally accommodated at the domain exit.

Walls: A no-slip boundary condition was applied to all solid wall surfaces in both simulations, enforcing zero velocity at the wall and allowing the boundary layer to develop naturally within the mesh resolution provided.

4.1.5 Mesh Generation

Mesh Generation Strategy

A structured hexahedral mesh was generated for the computational domain, consisting of **1,743,532 elements** and **1,787,422 nodes**. Hexahedral elements were selected due to their

superior alignment with the flow direction and reduced numerical diffusion compared to tetrahedral alternatives.

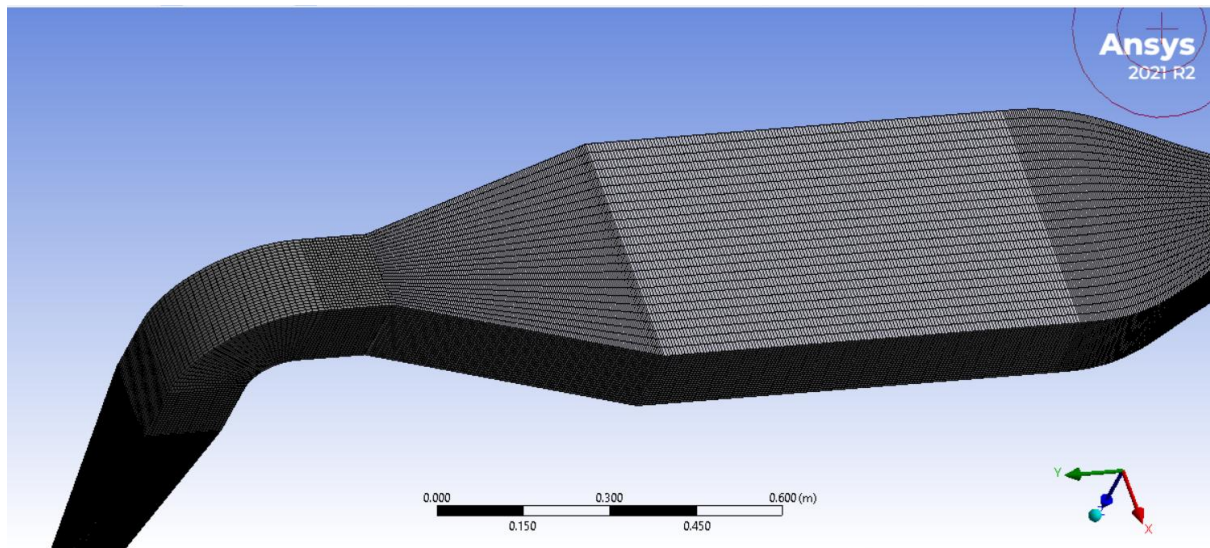


Figure 4.3 - Mesh

Mesh Quality analysis

The mesh quality was assessed using skewness and orthogonal quality metrics. The skewness of the majority of elements falls within the 0.00–0.06 range with no elements exceeding 0.80, and orthogonal quality peaks in the 0.95–1.00 range, confirming the mesh is of acceptable quality for the intended simulations.

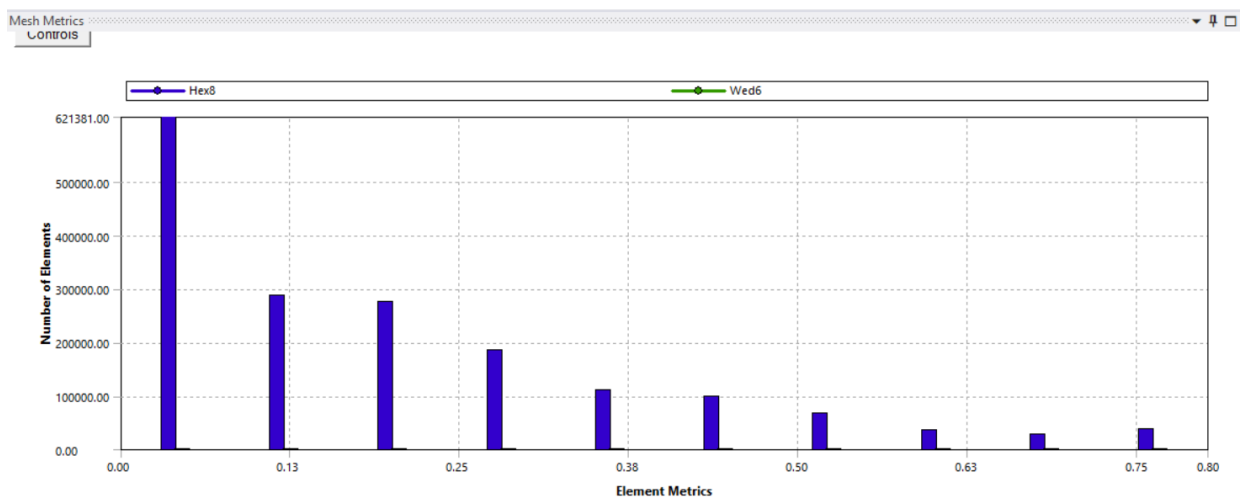


Figure 4.4 – Skewness

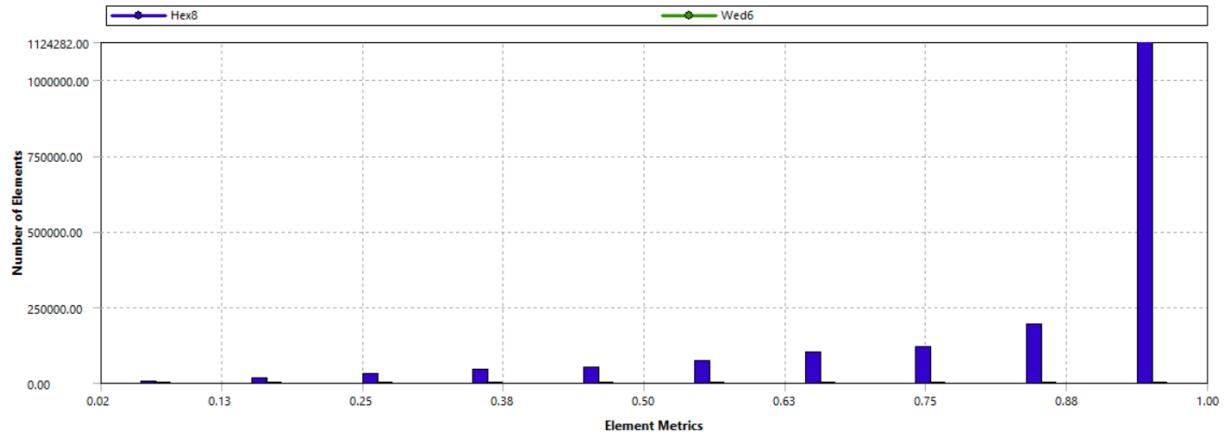


Figure 4.5 - Orthogonal quality

Near wall Treatment

Inflation layers were applied to all solid wall boundaries using the sweep and inflation method to resolve the viscous sublayer accurately. A first layer height of 5×10^{-5} m was specified, with 15 inflation layers and a growth rate of 1.2. This configuration ensures adequate near-wall mesh resolution consistent with the $y^+ \approx 1$ requirement of the SST $k-\omega$ turbulence model.

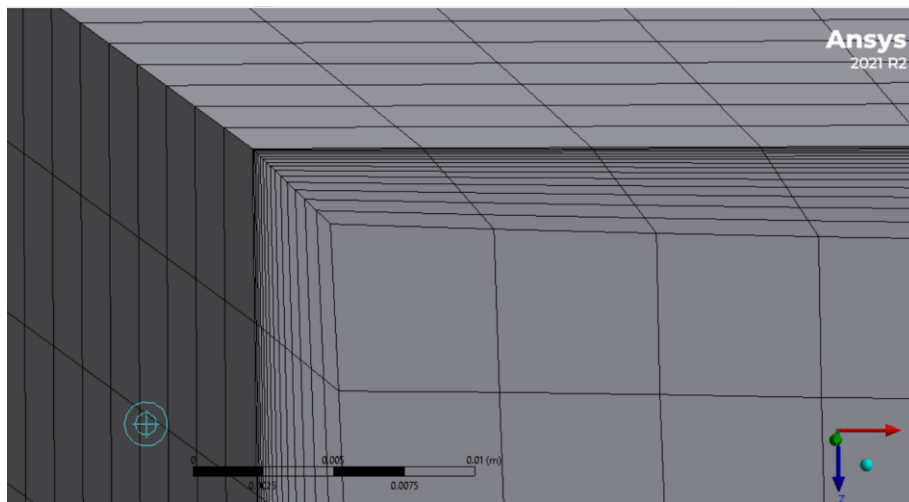


Figure 4.6 - Near wall treatment

Post-simulation inspection of the facet-average y^+ values confirmed values of 0.841, 0.897, and 1.136 across the principal wall boundaries, validating that the viscous sublayer was fully resolved throughout the domain.

4.2 Structural Analysis (FEA) Approach

4.2.1 Overview of Structural Analysis Methodology

Finite Element Analysis (FEA) was carried out using Abaqus/Standard to evaluate the structural adequacy of the glass fiber reinforced polymer (GFRP) components under internal water pressure loading. Three components were selected for analysis based on their large surface area and susceptibility to outward deflection under hydrostatic pressure: the contraction section, the wide-angle diffuser, and the settling chamber. The primary objective of the analysis was to determine whether the as-fabricated 3 mm GFRP wall thickness alone was sufficient to withstand the internal pressure without exceeding the material's tensile strength. The analysis was conducted in two stages a baseline simulation of the unreinforced shell, followed by a reinforced simulation incorporating external ribs to quantify the structural improvement achieved through reinforcement. Results of both stages are presented in Chapter 5.

4.2.2 Material Properties and Assumptions

All three components were fabricated using a hand-layup GFRP process with a total laminate of three layers: one layer of 300 GSM woven glass fiber fabric and two layers of 450 GSM woven glass fiber fabric, consolidated with polyester resin to a nominal wall thickness of 3 mm. The material was modelled as a linear elastic, homogeneous, and isotropic solid — a reasonable approximation for a balanced woven laminate under static loading conditions [21]. The material properties used in Abaqus, defined in the mm-tone-second (MTS) unit system, are given in Table 4.1.

Table 11: GFRP material properties used in Abaqus FEA (MTS unit system)

Property	Value	Unit
Density ρ	1.85×10^{-9}	tone/mm ³
Young's Modulus E	10,000	MPa
Poisson's Ratio ν	0.3	—
Tensile Strength σ_t	180	MPa
Wall Thickness t	3	mm

A tensile strength of 180 MPa was adopted based on published data for hand-layup GFRP laminates with similar fibre volume fractions of 35–40% [21], [3]. No failure criteria were explicitly defined within Abaqus; instead, the maximum Von Mises stress output from each simulation was compared against the tensile strength in the post-processing stage to assess structural adequacy.

4.2.3 Component Geometries and Element Type

The CAD geometry for each component was imported directly into Abaqus from the detailed design model. A mixed meshing strategy was adopted for all three components, using structured hexahedral elements on the regular wall regions and unstructured tetrahedral elements at the flange faces and geometrically complex transition regions where structured meshing was not feasible. The element types assigned to each region are summarized in Table 4.2.

Table 12: Element types assigned to each component in Abaqus

Component	Regular Wall Regions	Flange / Transition Regions
Contraction	C3D8R — 8-node linear brick, reduced integration	C3D10 — 10-node quadratic tetrahedron
Wide-angle diffuser	C3D20R — 20-node quadratic brick, reduced integration	C3D10 — 10-node quadratic tetrahedron
Settling chamber	C3D20R — 20-node quadratic brick, reduced integration	C3D10 — 10-node quadratic tetrahedron

The C3D8R element was selected for the contraction section wall regions due to the predominantly structured, regular geometry of the contracting surface. For the wide-angle diffuser and settling chamber, the higher-order C3D20R element was employed to better capture the stress gradients in these geometrically simpler but larger flat-panel regions, where bending-dominated deformation under pressure loading benefits from quadratic interpolation [22]. The C3D10 quadratic tetrahedral element was used consistently across all three components for the flange regions, providing accurate stress recovery at the constrained boundaries without requiring structured meshing. The meshes for all three components are shown in Figures 4.16, 4.17, and 4.18 respectively.

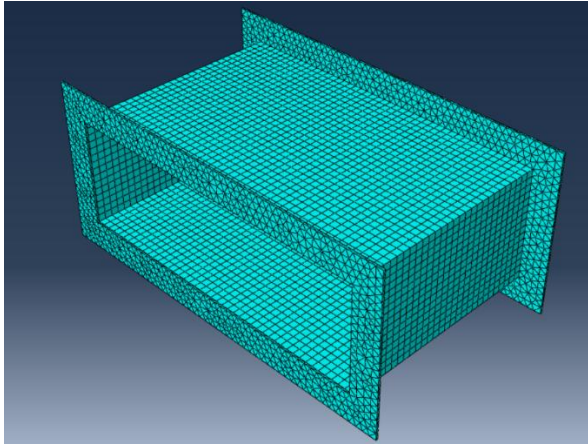


Figure 4.7: Abaqus mesh - settling chamber

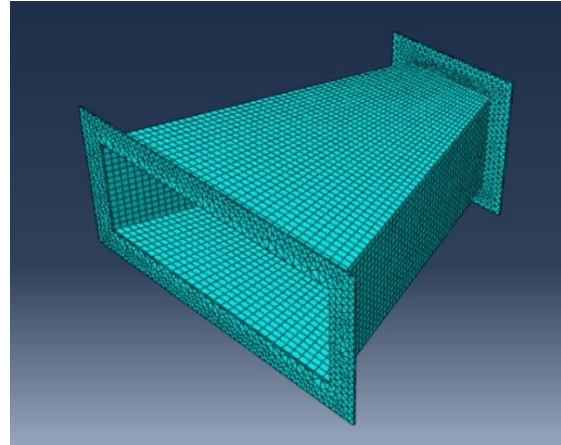


Figure 4.8: Abaqus mesh - wide angle diffuser

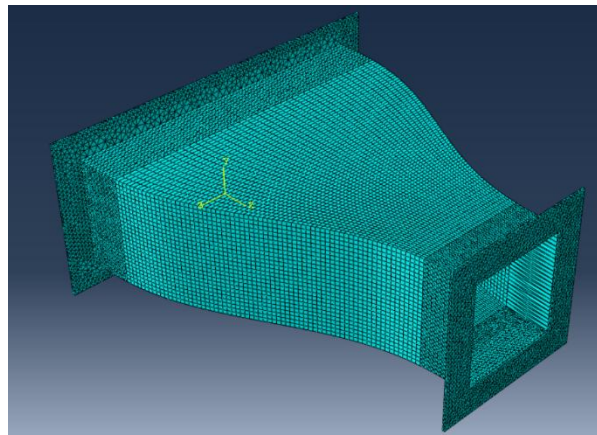


Figure 4.9: Abaqus mesh - contraction section

4.2.4 Boundary Conditions and Loading

Identical boundary conditions were applied to all three components, reflecting their physical installation condition within the tunnel circuit. Both the inlet and outlet flanges of each component were fully constrained in all three translational degrees of freedom ($U1 = U2 = U3 = 0$), representing the rigid connection to adjacent tunnel components at each flange face. Rotational degrees of freedom ($UR1, UR2, UR3$) were left unconstrained, consistent with a pinned-flange condition. A ramp amplitude was applied to both the boundary conditions and the pressure load to ensure numerical stability during the static solution. The boundary condition setup for settling chamber is shown in figure 4.19, which is similar to other two components as well.

A uniform internal pressure of **0.02 MPa (20 kPa)** was applied to all inner surfaces of each component, representing the hydrostatic pressure exerted by the internal water column. This

value corresponds to a water column height of approximately 2 m, consistent with the worst-case hydrostatic head in the vertical tunnel circuit:

$$p = \rho gh = 1000 \times 9.81 \times 2.0 \approx 19,620 \text{ Pa} \approx 0.02 \text{ MPa}$$

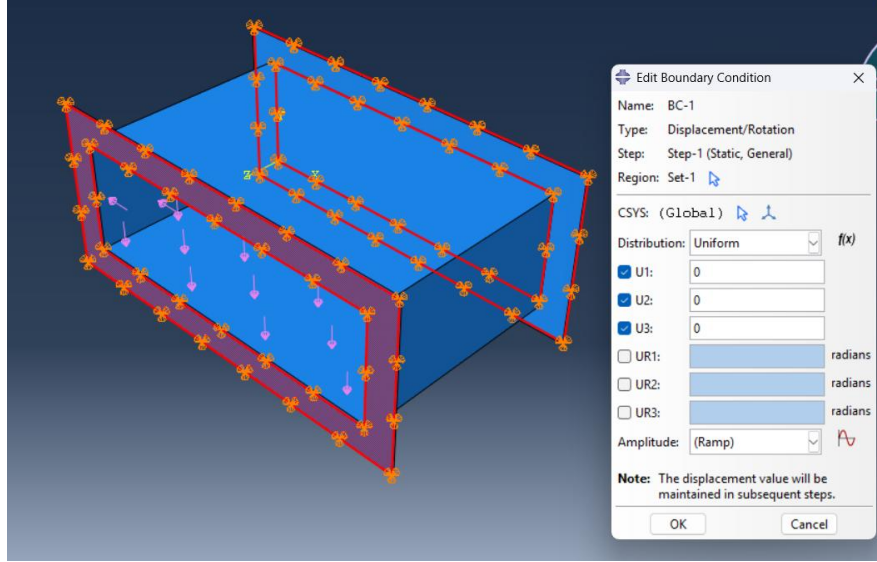


Figure 4.10: Boundary conditions - settling chamber (other components are also like this)

4.2.5 Analysis Procedure and Acceptance Criteria

All simulations were performed as Static, General steps in Abaqus/Standard, appropriate for linear elastic structural analysis under static pressure loading. The primary output quantities extracted from each simulation were:

- **Maximum Von Mises stress** σ_{VM} — used to assess the onset of yielding or failure
- **Maximum displacement** U_{max} — used to assess geometric distortion of the tunnel profile

The acceptance criterion for structural adequacy was defined as:

$$\sigma_{VM} \leq \frac{\sigma_t}{FoS} \quad (40)$$

where $\sigma_t = 180 \text{ MPa}$ is the tensile strength of the GFRP laminate and a factor of safety of $FoS = 3$ was applied, consistent with recommended practice for hand-layup composite structures subject to manufacturing variability [21]. This gives an allowable stress of:

$$\sigma_{allowable} = \frac{180}{3} = 60 \text{ MPa} \quad (41)$$

Where the baseline simulation indicated that σ_{VM} exceeded $\sigma_{allowable}$, external ribs were added to the component geometry and the simulation was repeated to verify that the reinforced design satisfied the acceptance criterion. The rib design and the results of both baseline and reinforced simulations for all three components are presented in Section 5.2.

4.3 Finalized Component Dimensions and Design Parameters

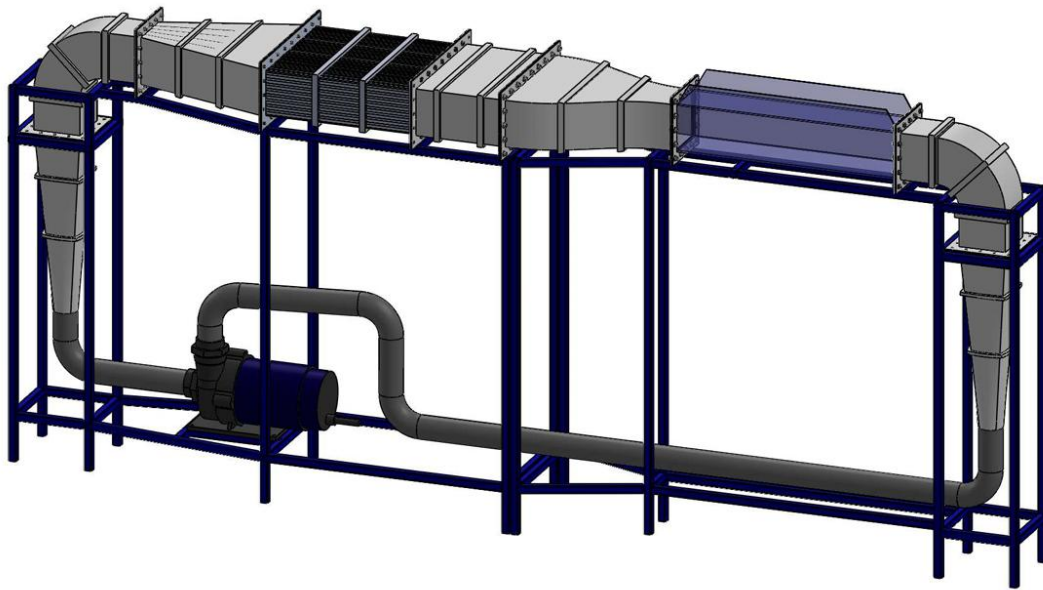


Figure 4.11: Complete CAD model of the Hydrodynamic tunnel

4.3.1 Vertical Transition section

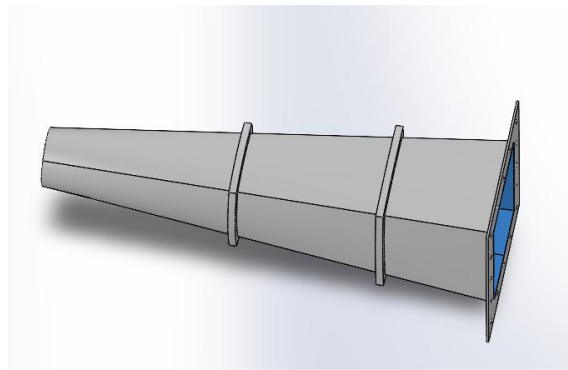


Figure 4.12: vertical transition section

The vertical transition section reduces pump-generated turbulence and compensates for gravitational pressure loss through pressure recovery. It's designed to gradually expand the flow from a circular inlet of 75 mm diameter to a square outlet of 150 mm × 150 mm while minimizing flow separation and pressure losses. The component was fabricated using GFRP hand layup with structural reinforcements to ensure smooth flow development and adequate rigidity.

Table 4.13: vertical transition section Detailed design parameters

Parameter	Value	Description
Geometry & Dimensions		
Overall length	613 mm	Along the central axis of the transition
Inlet diameter	75 mm	Circular cross-section at inlet
Outlet dimensions	150 mm × 150 mm	Square cross-section at outlet
Transition angle	7°	Total divergence angle
Material & Layup		
Material	GFRP	Woven glass fiber reinforced polymer
Layer 1	300 GSM × 1 layer	Inner surface layer for finish quality
Layers 2 & 3	450 GSM × 2 layers	Structural layers for load-bearing capacity
Resin system	Polyester resin	Compatible with hand layup process
Manufacturing Process		
Fabrication method	Hand layup	Manual wet layup of glass fiber plies
Mold type	Thin sheet metal mold	Forming surface for the transition geometry
Structural reinforcement	External wood frames	Outer wall reinforcement for added rigidity

4.3.2 Rectangular Bend

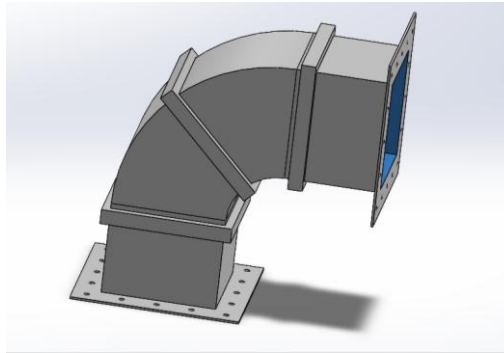


Figure 4.13: Rectangular 90-degree bend

The rectangular 90-degree bend is used to redirect the flow within the hydrodynamic tunnel while maintaining a consistent square cross-section throughout the section. Its primary purpose is to guide the flow smoothly around the corner with minimum pressure loss and reduced flow separation. The bend was fabricated using GFRP through a hand layup process.

Table 4.14: 90 degree bend Detailed design parameters

Parameter	Value	Notes
Geometry & Dimensions		
Inlet dimensions	150 mm × 150 mm	Square cross-section at inlet
Outlet dimensions	150 mm × 150 mm	Square cross-section at outlet
Bend angle	90°	Right-angle turn
Material & Layup		
Material	GFRP	Woven glass fiber reinforced polymer
Layer 1	300 GSM × 1 layer	Inner surface layer
Layers 2 & 3	450 GSM × 2 layers	Structural layers
Resin system	Polyester resin	Hand layup compatible
Manufacturing Process		
Fabrication method	Hand layup	Manual wet layup of glass fiber plies
Assembly method	Template-guided fixing	Flat GFRP panels aligned and fixed using templates; no formal mold used
Structural reinforcement	External wood frames	Outer wall rigidity and strength

4.3.3 Wide Angle Diffuser

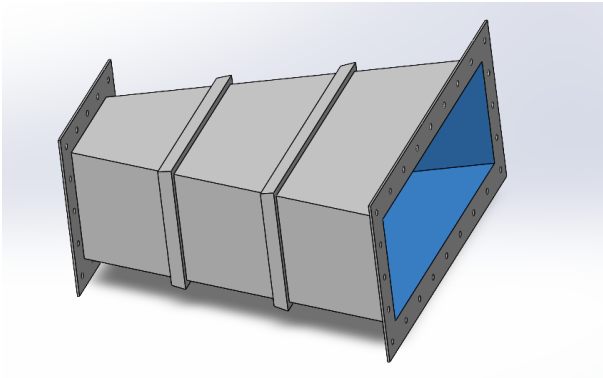


Figure 4.14: Wide angle diffuser

The wide-angle diffuser is used to gradually reduce the flow velocity and recover static pressure before the fluid enters the downstream sections. Due to the large divergence angle, four internal vanes were incorporated to guide the flow and minimize flow separation. The diffuser was fabricated using GFRP through a hand layup process.

Table 4.15: wide angle diffuser Detailed design parameters

Parameter	Value	Notes
Geometry & Dimensions		
Overall length	475 mm	Square cross-section at inlet
Total divergence angle	35°	Full included angle of the diffuser walls
Number of vanes	4	Flow-straightening vanes to control separation
Vane angle	7°	Individual vane setting angle relative to flow axis
Material & Layup		
Material	GFRP	Woven glass fiber reinforced polymer
Layer 1	300 GSM × 1 layer	Inner surface layer
Layers 2 & 3	450 GSM × 2 layers	Structural layers
Resin system	Polyester resin	Hand layup compatible
Manufacturing Process		
Fabrication method	Hand layup	Manual wet layup of glass fiber plies
Assembly method	Template-guided fixing	Flat GFRP panels aligned and fixed using templates; no formal mold used
Structural reinforcement	External wood frames	Outer wall rigidity and strength

4.3.4 Honeycomb section

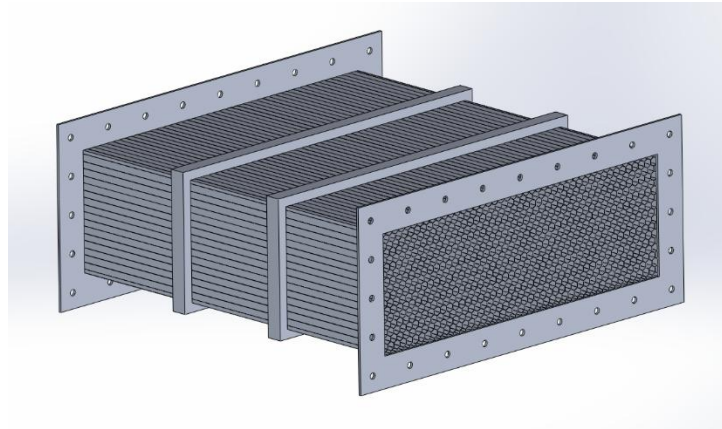


Figure 4.15: Honeycomb structure

The honeycomb section is installed to reduce large-scale turbulence and remove swirl generated by upstream components such as the pump and bends. Its hexagonal cell structure guides the flow into parallel passages, improving flow uniformity before the contraction and test section. The honeycomb was manually fabricated using thin aluminum foil sheets bent and bonded together to form the required geometry. Wooden and glass fiber supports were used to provide structural stability and secure the component within the tunnel.

Table 4.16: Honeycomb structure detailed design parameters

Parameter	Value	Description
Geometry & Dimensions		
Honeycomb type	Hexagonal cell structure	Selected for improved flow uniformity and minimal gaps
Overall length	500 mm	Length of the honeycomb section along flow direction
Cell diameter	10 mm	Hydraulic diameter of individual honeycomb cells
Cross-sectional size	450 mm × 150 mm	Matches settling chamber dimensions
Material & Construction		
Cell material	Aluminum foil	Lightweight material for flow conditioning
Structure type	Honeycomb array	Parallel flow passages for swirl reduction
Wall thickness	0.1–0.2 mm	Thin walls to minimize pressure loss
Support frame	Wooden and glass fiber	Provides structural stability inside chamber
Manufacturing Process		

Fabrication method	Aluminum foil sheets were bent and bonded together to form a hexagonal honeycomb structure	Manual fabrication process used to achieve the required cell geometry
Installation location	Settling chamber	Positioned upstream of the contraction cone
Mounting method	Bolted / fixed support	Secures honeycomb within tunnel structure

4.3.5 Settling chamber

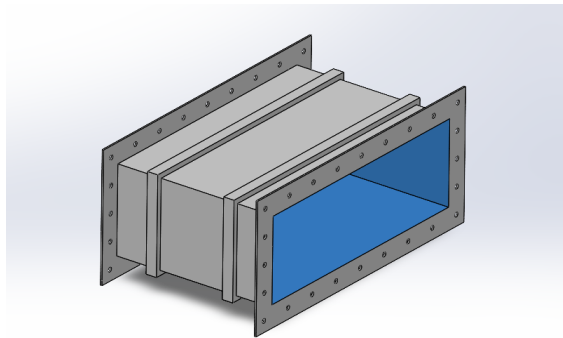


Figure 4.16: Settling chamber

The settling chamber is used to stabilize and condition the flow before it enters the contraction and test section. Its primary purpose is to reduce turbulence intensity and provide a more uniform velocity distribution by allowing disturbed flow structures to dissipate. The chamber was fabricated using GFRP through a hand layup process, providing sufficient structural strength and a smooth internal surface.

Table 4.17: settling chamber Detailed design parameters

Parameter	Value	Description
Geometry & Dimensions		
Overall length	300 mm	Along the central axis
Material & Layup		
Material	GFRP	Woven glass fiber reinforced polymer
Layer 1	300 GSM $\times$ 1 layer	Inner surface layer
Layers 2 & 3	450 GSM $\times$ 2 layers	Structural layers
Resin system	Polyester resin	Hand layup compatible
Manufacturing Process		
Fabrication method	Hand layup	Manual wet layup of glass fiber plies

Structural reinforcement	External wood frames	Outer wall rigidity and strength
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4.3.6 Contraction section

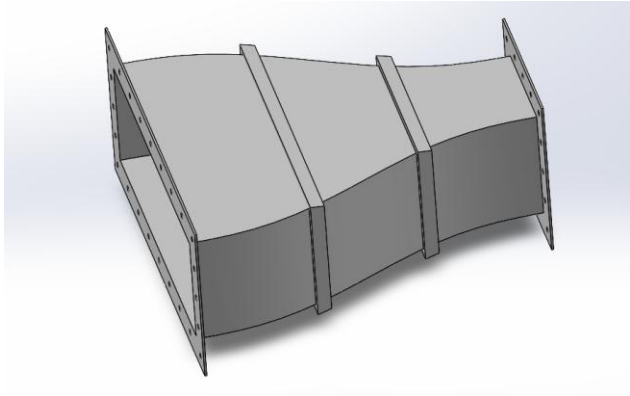


Figure 4.17: Contraction section detailed design

The contraction section is used to accelerate the flow and improve flow uniformity before the fluid enters the test section of the hydrodynamic tunnel. By gradually reducing the cross-sectional area, it helps decrease turbulence intensity and produce a smoother, more stable flow profile. The component was fabricated using GFRP through a hand layup process with multiple glass fiber layers to provide structural strength and a smooth internal surface.

Table 4.18: Contraction section Detailed design parameters

Parameter	Value	Notes
Geometry & Dimensions		
Overall length	522 mm	Along the central axis of the contraction
Material & Layup		
Material	GFRP	Woven glass fiber reinforced polymer
Layer 1	300 GSM × 1 layer	Inner surface layer
Layers 2 & 3	450 GSM × 2 layers	Structural layers
Resin system	Polyester resin	Hand layup compatible
Manufacturing Process		
Fabrication method	Hand layup	Manual wet layup of glass fiber plies
Structural reinforcement	External wood frames	Outer wall rigidity and strength

4.3.7 Test Section

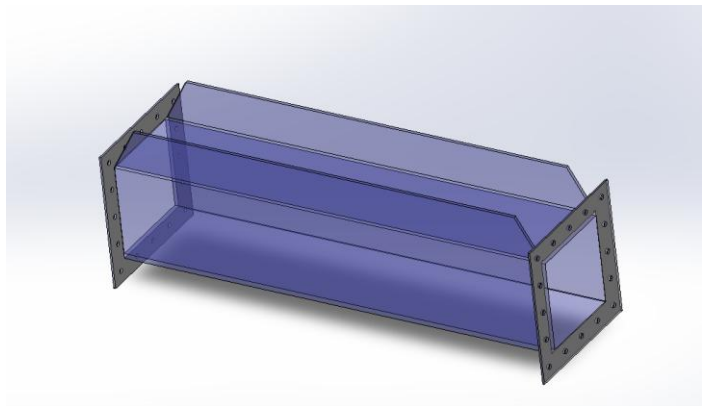


Figure 4.18: Test section detailed drawing

The test section serves as the primary experimental region where flow conditions are evaluated under controlled and observable conditions. Its main purpose is to provide a uniform and stable flow field after the contraction section, ensuring that measurements are not influenced by upstream disturbances. It also enables detailed flow visualization and data acquisition, allowing assessment of velocity distribution, turbulence characteristics, and overall flow quality. The transparent acrylic construction further supports optical access, making it suitable for visualization studies.

Table 4.19: Test section Detailed design parameters

Parameter	Value	Notes
Cross-section	150 mm × 150 mm	Square open channel cross-section
Overall length	750 mm	Along the central axis of the test section
Channel type	Open channel	Top surface open to atmosphere
Material	Transparent acrylic	Allows full optical access for flow visualization

5 Chapter 4 - Results and Discussion

5.1 Analytical Calculation Results *[to be updated after verification]*

5.1.1 Head Loss Analysis Results

5.1.2 Pump Adequacy Verification

5.2 Simulation Results - CFD

5.2.1 Wide-Angle Diffuser - Baseline Simulation

Transient Flow Separation Analysis

The transient simulation was carried out to capture the unsteady flow behavior within the wide-angle diffuser. Unlike the steady-state approach, the transient simulation resolves the time-dependent development of flow separation, revealing the location, extent, and evolution of separated flow regions over time.

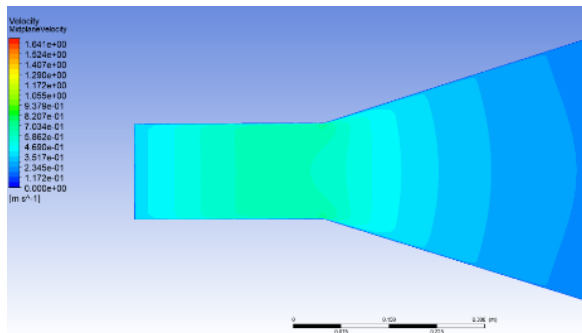


Figure 5.1: 0.25 S - Early transient stage (Attached flow)

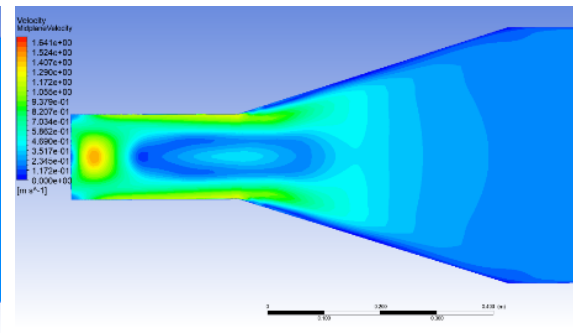


Figure 5.2: 1.5 S – Separation forming

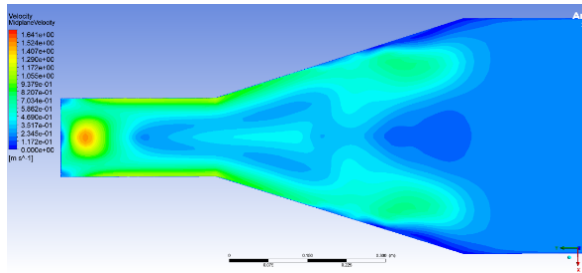


Figure 5.3: 2.2 S - Developed separation

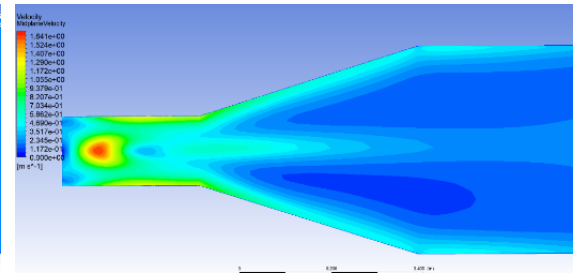


Figure 5.4: Timestep 19.6 S– Quasi steady flow

The transient velocity contours in Figures 5.1 through 5.4 show how the flow inside the wide-angle diffuser changes over time. At $t = 0.25$ s, the flow is smooth and moves evenly through the diffuser without any issues. At $t = 1.5$ s, the flow begins to separate from the walls and a recirculation bubble starts to form. By $t = 2.2$ s, the separation is fully developed and the flow splits into two high-velocity regions along the top and bottom walls, with a large slow-moving zone forming in the center. At $t = 2.8$ s, this central dead zone grows significantly larger while the two wall jets persist, maintaining a symmetric but severely separated flow pattern. By $t = 19.6$ s, the flow settles into a quasi-steady state where the two wall jets remain but become unequal in strength, and the central dead zone dominates most of the diffuser cross-section. Throughout this process, the separated flow forces the

remaining fluid into a high-velocity jet through the core of the diffuser, retaining kinetic energy that should have been converted into static pressure recovery. This represents a direct loss in diffuser efficiency. These results confirm that the wide-angle diffuser without vanes suffers from severe flow separation that persists over time. This uneven flow would carry downstream and negatively affect the quality of flow in the test section. Therefore, splitter vanes were introduced into the diffuser to divide the flow into smaller channels, reducing the effective expansion angle and suppressing flow separation.

Pressure Recovery Analysis at Wide angle diffuser

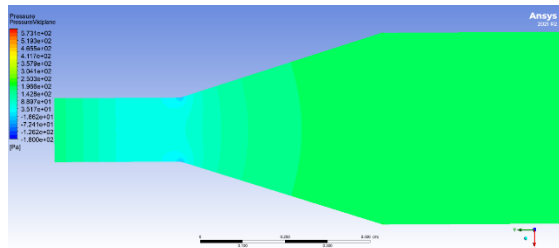


Figure 5.5: Timestep 25 - Uniform Pressure Recovery

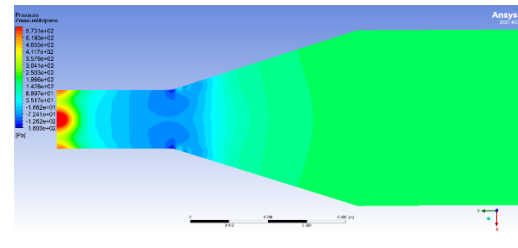


Figure 5.6: Timestep 150 - Early pressure disturbance

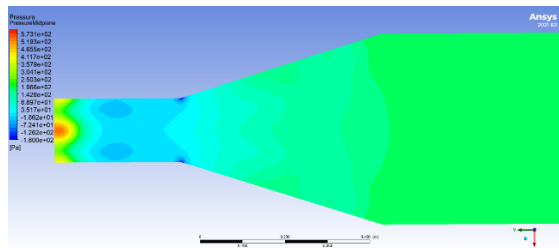


Figure 5.7: Timestep 220 - Symmetric Low-Pressure Regions

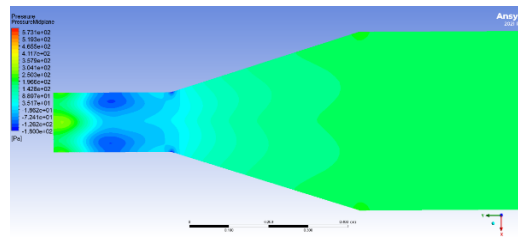


Figure 5.8: Timestep 1960 - Nearly stable pressure distribution

The transient pressure contours in Figures 5.5 through 5.8 show how pressure changes inside the wide-angle diffuser over time. At $t = 0.25$ s, the pressure rises smoothly from inlet to exit, showing good pressure recovery. At $t = 1.5$ s, negative pressure zones appear in the upstream region as the recirculation bubble develops, and a high-pressure spot forms at the inlet wall. By $t = 2.2$ s, two low-pressure pockets form symmetrically about the centerline, matching the split flow pattern seen in the velocity contours. At $t = 2.8$ s, these low-pressure zones grow and extend further into the diffuser. By $t = 19.6$ s, several alternating low-pressure zones settle into a nearly stable pattern. Throughout all time steps, the diffuser exit maintains a relatively uniform positive pressure, confirming that some pressure recovery is

still achieved. However, the highly disturbed pressure field inside the diffuser confirms that significant pressure recovery is lost due to flow separation.

Pressure Recovery Coefficient at Quasi-Steady State

The Pressure Recovery Coefficient (C_P) at quasi-steady state was evaluated using the expression:

$$C_P = \frac{P_{out} - P_{in}}{0.5 \times \rho \times V_{in}^2} \quad (42)$$

where P_{out} and P_{in} are the area-averaged static pressures at the diffuser outlet and inlet respectively, ρ is the fluid density of water (998.2 kg/m³), and V_{in} is the mean inlet velocity at the diffuser inlet (m/s). The pressure recovery coefficient quantifies the fraction of dynamic pressure converted to static pressure recovery along the diffuser. The computed value of C_P at quasi-steady state was found to be 0.593, indicating that approximately 59.3% of the inlet dynamic pressure was successfully recovered as static pressure through the diffuser section.

5.2.2 Turbulence Intensity Decay Analysis

Turbulence intensity is a key parameter governing the flow quality within the test section of a hydrodynamic tunnel. High turbulence levels in the flow can significantly affect experimental measurements by introducing unsteady fluctuations that interfere with the behavior of test models. Therefore, understanding how turbulence evolves through the tunnel components is essential for evaluating the suitability of the design for experimental testing. In this study, the inlet turbulence intensity was set to 8% at the vertical transition diffuser inlet, representing the turbulent flow delivered by the recirculation pump.

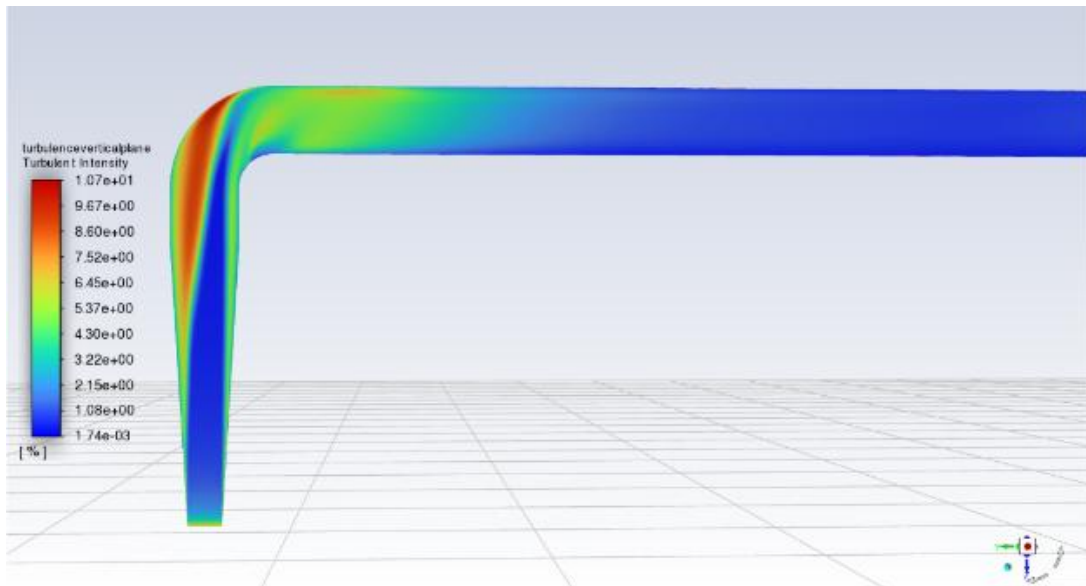


Figure 5.9 - Turbulence intensity variation from diffuser inlet

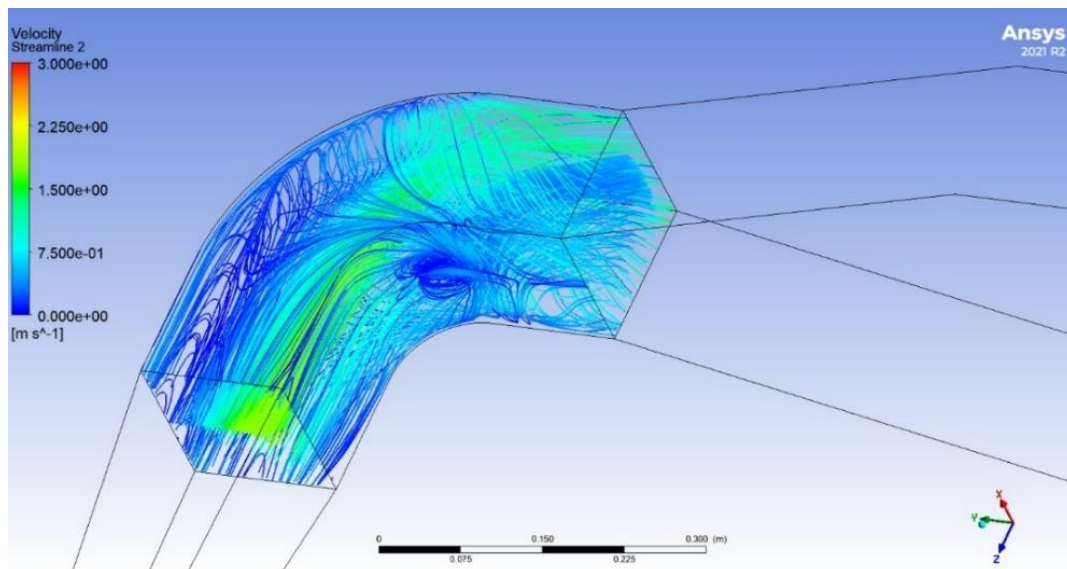


Figure 5.10 - 90 bend Velocity streamlines

Figure 5.9 shows the turbulence intensity distribution along the vertical transition diffuser and 90° bend on a vertical plane. In the original design intent, the vertical diffuser was included to provide static pressure rise and to allow turbulence intensity to decay before the flow enters the wide-angle diffuser. However, the simulation revealed that while the vertical diffuser does reduce the turbulence intensity as intended, the 90° bend immediately following it reintroduces turbulence back into the flow. As clearly visible in Figure 5.9, the turbulence intensity is relatively low within the vertical diffuser section, but rises sharply to a peak of 10.71% at the outer wall of the 90° bend. This increase is caused by the strong

flow curvature and secondary flow effects induced by the bend geometry, which energize the boundary layer and generate additional turbulent fluctuations. This finding highlights that the 90° bend acts as an additional turbulence source beyond the initial design assumption.

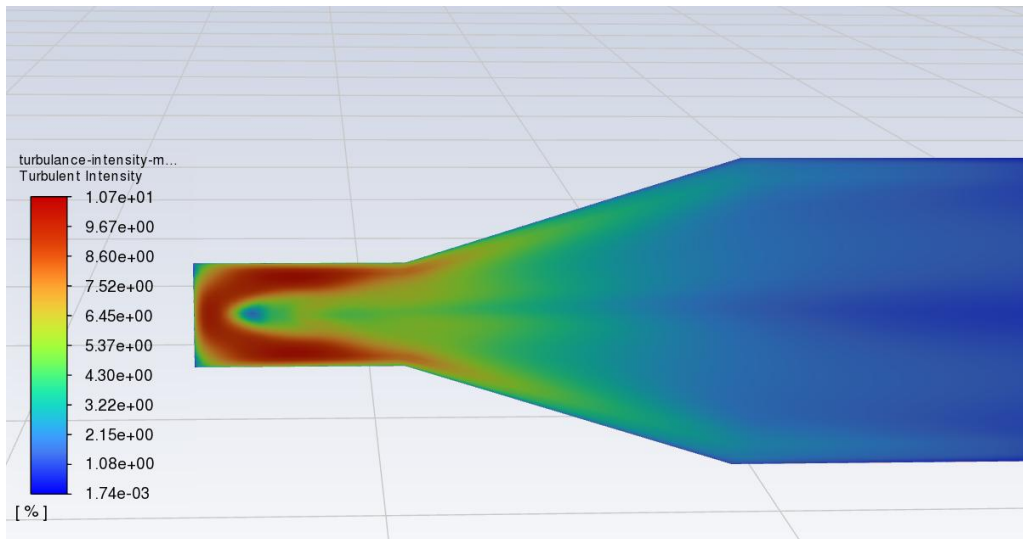


Figure 5.11 - Turbulence intensity decay through the wide-angle diffuser

Figure 5.11 shows the turbulence intensity distribution on the horizontal midplane, capturing the flow through the wide-angle diffuser. As the high turbulence flow entering the wide-angle diffuser at 10.71% passes through the expanding diffuser section, the turbulence intensity decays significantly along the diffuser length. This decay occurs due to the increasing cross-sectional area of the diffuser, which reduces the mean flow velocity and allows the turbulent fluctuations to dissipate. By the diffuser exit, the turbulence intensity has reduced to approximately 2.08%, representing a significant improvement in flow quality. Following the diffuser exit, the flow passes through the honeycomb flow straightener, which is expected to reduce the turbulence intensity even further by breaking down the remaining large-scale turbulent structures before the flow enters the settling chamber and test section. This confirms that the wide-angle diffuser, despite suffering from flow separation in its baseline configuration, still plays an important role in decaying turbulence intensity along the tunnel circuit.

5.2.3 Design Modifications Arising from CFD

The CFD simulation of the baseline wide-angle diffuser configuration clearly demonstrated that the diffuser in its original form suffers from severe flow separation, poor pressure recovery, and uneven flow conditions. These findings established the need for a design

modification to suppress flow separation and improve the overall flow quality delivered to the settling chamber and test section. Based on the simulation findings and supported by the work of Waitman, Reneau, and Kline [12], who investigated the effects of inlet conditions on the performance of two-dimensional subsonic diffusers, splitter vanes were introduced into the wide-angle diffuser as the primary design modification.

Splitter Vane Design and Parameters

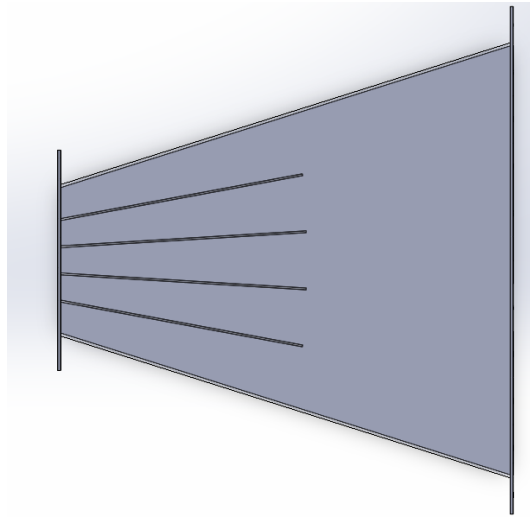


Figure 5.12 - Splitter vane design

The splitter vane geometry was determined based on the design criteria established by Waitman, Reneau, and Kline [12]. Table 2, reproduced from their work, presents the recommended vane configurations for a range of diffuser divergence angles. For the present design with a total divergence angle of $2\theta = 35^\circ$, the highlighted row recommends a configuration of 4 vanes with an individual vane angle of $\alpha = 7^\circ$ relative to the centerline, which was adopted as the basis for the splitter vane design in the present study.

2 θ , deg	n	α , deg	b/a	b	a	f/a_{max}	fmax	ftested
6	1	3	1.2	1.8	1.5	11.4	17.1	15
10	1	5	1.2	1.8	1.5	11.4	17.1	15
14	1	7	1.2	1.8	1.5	11.4	17.1	15
21	2	7	1.2	1.2	1	11.4	11.4	12
28	3	7	1.2	0.9	0.75	11.4	8.56	9

35	4	7	1.2	0.72	0.6	11.4	6.84	6
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Table 20 - Waitman, Reneau and Kline Recommendations

Based on these recommendations and the physical dimensions of the wide-angle diffuser, the actual vane parameters were calculated and are summarized in Table 1. The spacing between adjacent vanes and between the wall and the adjacent vane were determined to closely match the recommended ratios from the reference design criteria. The vane length in the flow direction was also selected to fall within the recommended range, ensuring effective flow guidance through each sub-channel.

Parameter	Description	Value
Number of vanes	Total vanes installed	4
α	Vane angle relative to centerline	7°
t	Vane thickness	2mm
a	Minimum spacing between adjacent vanes	25.98 mm
b	Minimum spacing between wall and adjacent vane	31.98 mm
b/a	Spacing ratio	1.23
f	Vane length in flow direction	260 mm
f/a	Vane aspect ratio	10

Table 21 - Splitter vane parameters

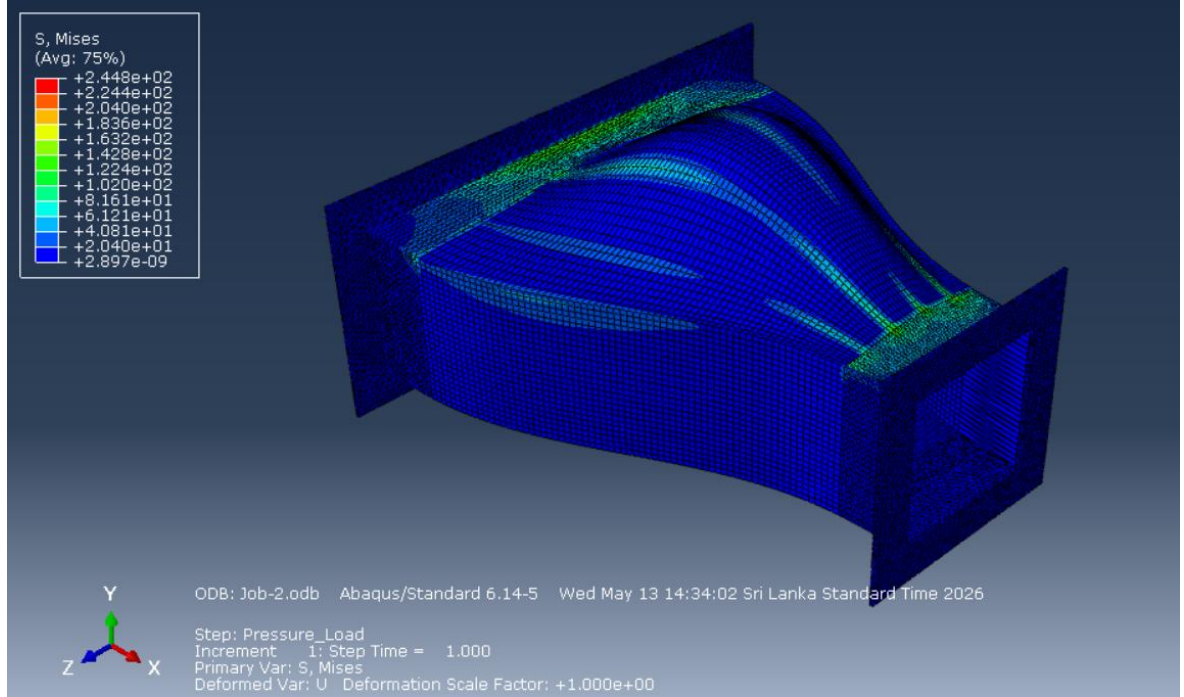
5.3 Simulation results - FEA

Baseline FEA simulations were conducted for the contraction section, wide-angle diffuser, and settling chamber under a uniform internal pressure of 0.02 MPa, with no reinforcement present. The objective was to determine whether the as-designed 3 mm GFRP wall could independently withstand the hydrostatic loading, or whether additional structural reinforcement was required. The acceptance criterion, as defined in Section 4.2.5, requires:

$$\sigma_{VM} \leq \sigma_{allowable} = \frac{\sigma_t}{FoS} = \frac{180}{3} = 60 \text{ MPa} \quad (43)$$

The Von Mises stress distributions for all three components are presented below.

5.3.1 Contraction Section – baseline FEA results analysis



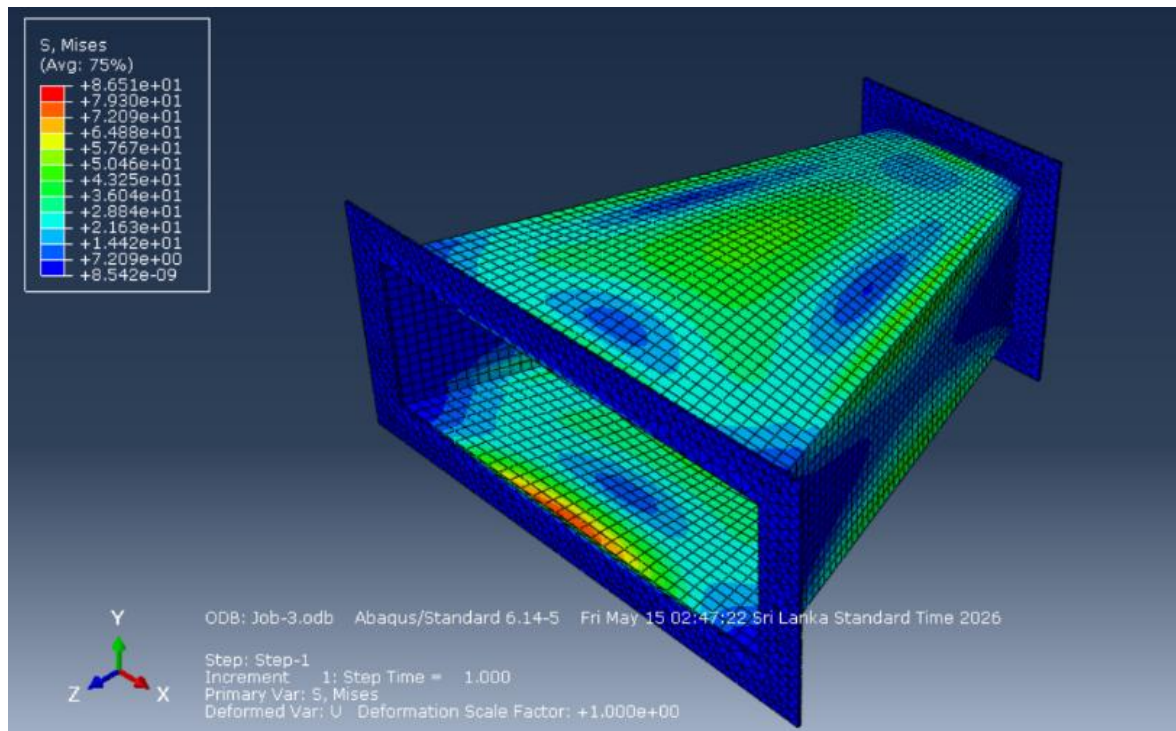
The baseline simulation of the contraction section produced a maximum Von Mises stress of **244.8 MPa**, which not only exceeds the allowable stress of 60 MPa but critically exceeds the tensile strength of the GFRP material (180 MPa). This indicates that the unreinforced 3 mm wall would undergo catastrophic failure under the design pressure loading. The highest stress concentrations are observed along the curved wall regions of the Bell-Mehta contraction profile, where the non-uniform curvature generates localized bending and membrane stress that the thin shell cannot resist without support. The flanges, being fully constrained, show near-zero stress as expected.

The large discrepancy between the applied pressure (0.02 MPa) and the resulting stress level is explained by the thin-shell bending mechanics governing the response. For a thin curved plate under uniform pressure, the bending stress scales as:

$$\sigma \propto \frac{p \cdot a^2}{t^2} \quad (44)$$

where a is the unsupported span and t is the wall thickness. With $t = 3$ mm and large unsupported curved spans across the contraction profile, the resulting bending stress amplification is severe, producing stresses far exceeding what the membrane alone would suggest [23].

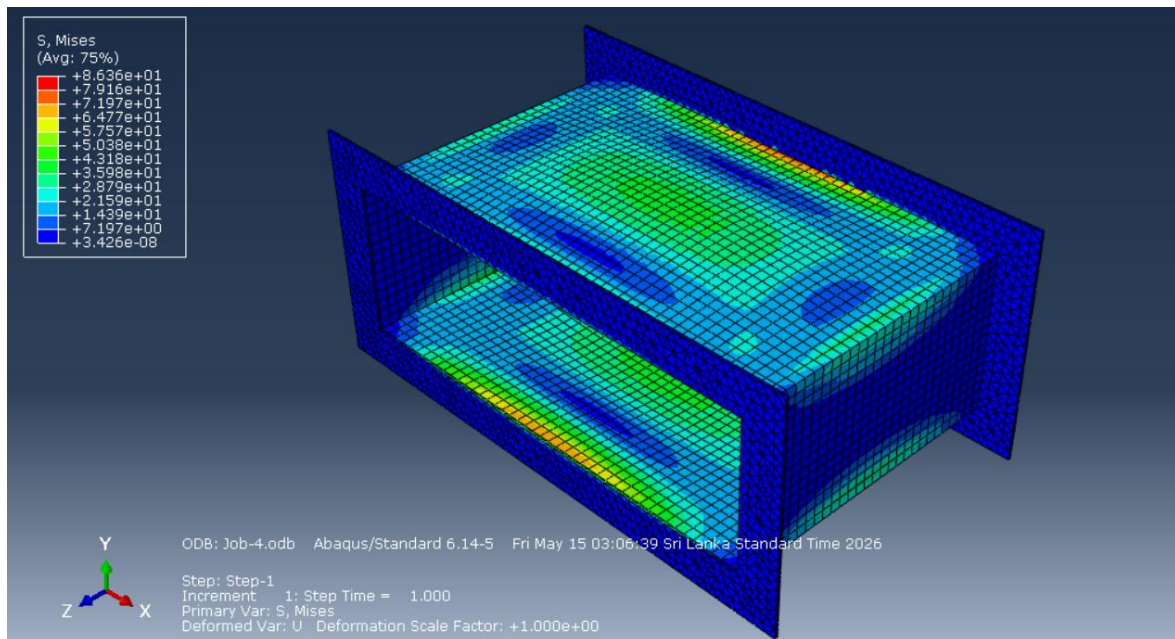
5.3.2 Wide-Angle Diffuser - baseline FEA results analysis



The baseline simulation of the wide-angle diffuser yielded a maximum Von Mises stress of **86.51 MPa**, exceeding the allowable stress of 60 MPa but remaining below the tensile strength of 180 MPa. The stress distribution shows that the majority of the component operates at low stress levels (below 22 MPa), with the critical concentration localized at the mid-span of the bottom wall, where the longest unsupported flat panel span produces the highest bending moment under the applied pressure. This behavior is consistent with classical plate bending theory for a uniformly loaded rectangular plate with fixed edges, where maximum stress occurs at the center of the longest free span [23].

While the component would not undergo immediate fracture at this stress level, the exceedance of the allowable stress with a factor of safety of 3 indicates that the design does not provide adequate structural reliability for sustained operation under the internal water column pressure.

5.3.3 Settling Chamber - baseline FEA results analysis



The baseline simulation of the settling chamber produced a maximum Von Mises stress of **86.36 MPa**, almost identical to the wide-angle diffuser result. The stress distribution is similarly characterized by low stress across most of the component, with concentrated bending stress at the mid-span regions of the bottom and side walls. The near-symmetry of the stress pattern reflects the uniform pressure loading on a regular rectangular geometry. As with the wide-angle diffuser, the peak stress exceeds the allowable of 60 MPa, indicating structural inadequacy of the unreinforced 3 mm GFRP wall.

5.3.4 Summary of Baseline FEA Results

The results for all three components are summarized in Table 5.2.

Table 22: Summary of baseline FEA results for all three components

Component	Max σ_{VM} (MPa)	$\sigma_{allowable}$ (MPa)	$\sigma_{tensile}$ (MPa)	Status
Contraction section	244.8	60	180	X Exceeds tensile strength
Wide-angle diffuser	86.51	60	180	X Exceeds allowable
Settling chamber	86.36	60	180	X Exceeds allowable

All three components fail the structural acceptance criterion under the design internal pressure of 0.02 MPa. The contraction section represents the most critical case, with the

maximum Von Mises stress exceeding the tensile strength of the material, indicating that catastrophic failure would occur without reinforcement. The wide-angle diffuser and settling chamber, while not predicted to fracture, exhibit stresses 44% above the allowable limit, providing insufficient safety margin for reliable operation.

These findings confirm that the 3 mm GFRP wall thickness alone is structurally inadequate, and that additional reinforcement is necessary for all fabricated components. Based on these results, external ribs and additional GFRP reinforcement layers were incorporated into the physical fabrication of all three components, as described in Section 5.3. Post-reinforcement FEA simulations to verify the structural adequacy of the reinforced design are planned and will be presented in the revised submission.

5.4 Fabrication

5.4.1 Material Selection - Evaluation of Options & Justification of Glass Fiber

Table 23: Material selection criteria

Component	Selected Material	Alternative Considered	Reason for Selection over Alternative
Vertical diffusers, 90° bends, wide angle diffuser, settling chamber, contraction section	Glass fiber	Aluminum, Sheet metal	Corrosion resistant, lightweight, smooth internal surface, easy to fabricate complex shapes, lower cost
Test section	Acrylic	Glass	Optically transparent for flow visualization, lighter than glass, easier to machine and handle
Honeycomb	Aluminum sheets	Polycarbonate	Both equally suitable, polycarbonate unavailable in required specification

5.4.2 Fabrication Method and Process Overview

Glass Fiber Component Fabrication

For all glass fiber components, the hand layup method was employed. For all components except the vertical diffusers, flat glass fiber sheets were first manufactured using this method. To achieve a smooth internal surface finish, the glass fiber layers were laid up directly onto a flat glass surface, which acted as a mold. The laminate consisted of one 300 GSM glass fiber sheet sandwiched between two 450 GSM sheets, bonded using polyester resin and cured at room temperature. The cured sheets were then cut to the required part shapes using CNC machining. The individual cut parts were subsequently assembled and joined together to form each component, and additional glass fiber layers were applied at the edges and joints to provide structural strength and prevent leakage.



Figure 5.13: Glass fiber sheets fabrication



Figure 5.14 - CNC machined glass fiber sheet parts

5.4.3 Fabricated Components

Vertical Diffusers (×2)

The vertical diffusers are circular to rectangular transition sections connecting the circular pump outlet to the rectangular tunnel circuit. Due to the complex transition geometry, fabricating these components from flat glass fiber sheets was not feasible. Instead, a thin sheet metal mold was first constructed by bending and spot welding, and glass fiber layers were then laid up over this mold using the hand layup method with polyester resin and cured at room temperature.



Figure 5.15 - Thin sheet metal molds for vertical diffuser sections



Figure 5.16 - Vertical transition sections manufacturing

90° Bends (×2)

The 90° bends were fabricated using the same glass fiber sheet method described in Section 5.2.1. The cured glass fiber sheets were cut to the required part shapes using CNC machining and then assembled into the final bend geometry. To ensure precise alignment of the individual parts during assembly, laser cut plywood templates were used as alignment guides. Additional glass fiber layers were applied at the edges and joints after assembly to provide structural strength and prevent leakage.



Figure 5.17 - Plywood templates for part alignment during assembly



Figure 5.18 - Fabricated 90 bends

Wide-Angle Diffuser with Integrated Vanes

The wide-angle diffuser was fabricated using the same method as the previously described glass fiber components, where CNC machined glass fiber sheet parts were assembled and

reinforced with additional fiber layers at the edges and joints. The splitter vanes were designed to be fabricated from 2mm acrylic sheet due to its smooth surface finish and ease of cutting. However, the installation of the vanes into the diffuser body had not been completed at the time of writing and remains as part of the remaining fabrication work.

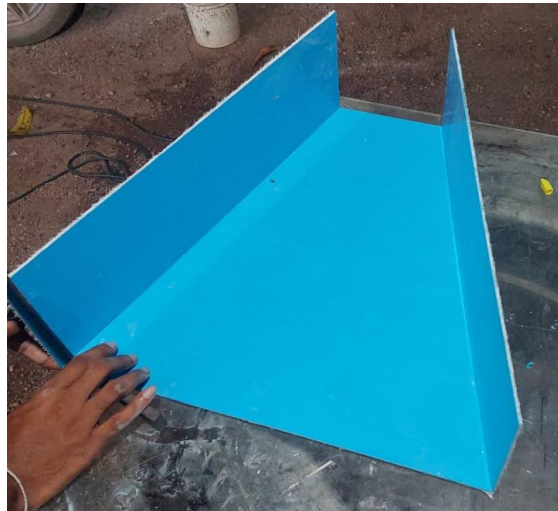


Figure 5.19 - Wide angle diffuser fabrication

Settling Chamber

The settling chamber was fabricated using the same method as the previously described glass fiber components, where CNC machined glass fiber sheet parts were assembled and reinforced with additional fiber layers at the edges and joints. Similar to the 90° bends, laser cut plywood templates were used during assembly to ensure precise alignment of the individual parts.

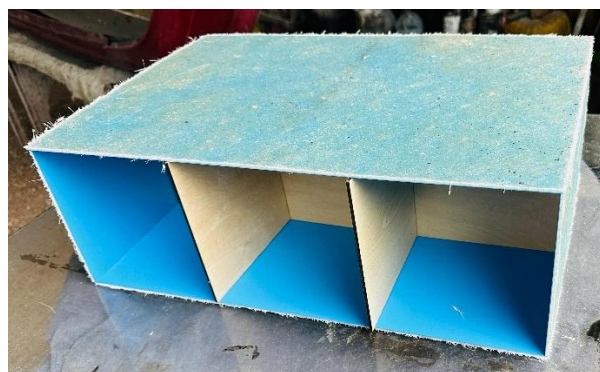


Figure 5.20 - Dimension verification using Plywood templates

Contraction Section

The contraction section was fabricated using the same method as the previously described glass fiber components, where CNC machined glass fiber sheet parts were assembled and

reinforced with additional fiber layers at the edges and joints. Laser cut plywood templates were used during assembly to ensure precise alignment of the individual parts. Due to the curved geometry of the contraction profile, achieving the correct curvature during assembly was particularly challenging. Additional plywood templates were therefore used to guide and maintain the curved shape accurately during the assembly process.

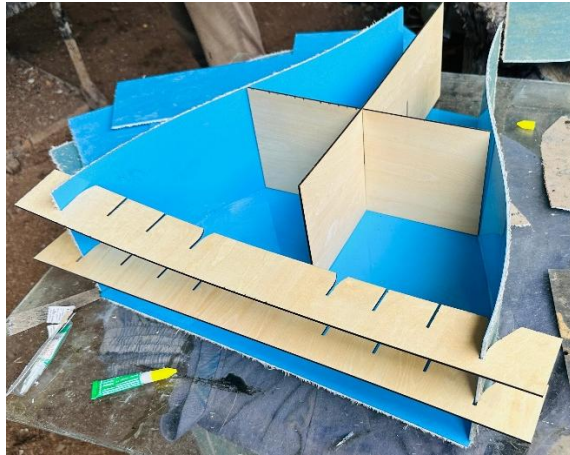


Figure 5.21- Plywood templates used to maintain curved contraction profile during assembly



Figure 5.22 - Contraction section

Test Section

The test section is fabricated from transparent acrylic sheets, selected for its optical transparency which is essential for flow visualization during experimental testing. The fabrication of the test section was ongoing at the time of writing.

Honeycomb Structure (Self-Fabricated) *[to be updated]*

5.4.4 Dimensional Verification and Quality Assessment *[to be updated June 19]*

5.5 Analysis of Hydrodynamic Tunnel Flow Quality *[to be updated June 19]*

5.6 Flow Visualization *[to be updated after testing]*

6 Chapter 5 - Conclusions and Recommendations *[to be updated after testing]*

6.1 Summary of Work Completed Against Objectives

6.2 Key Findings and Contributions

6.3 Limitations and Self-Critical Evaluation

6.4 Recommendations for Future Work

7 References

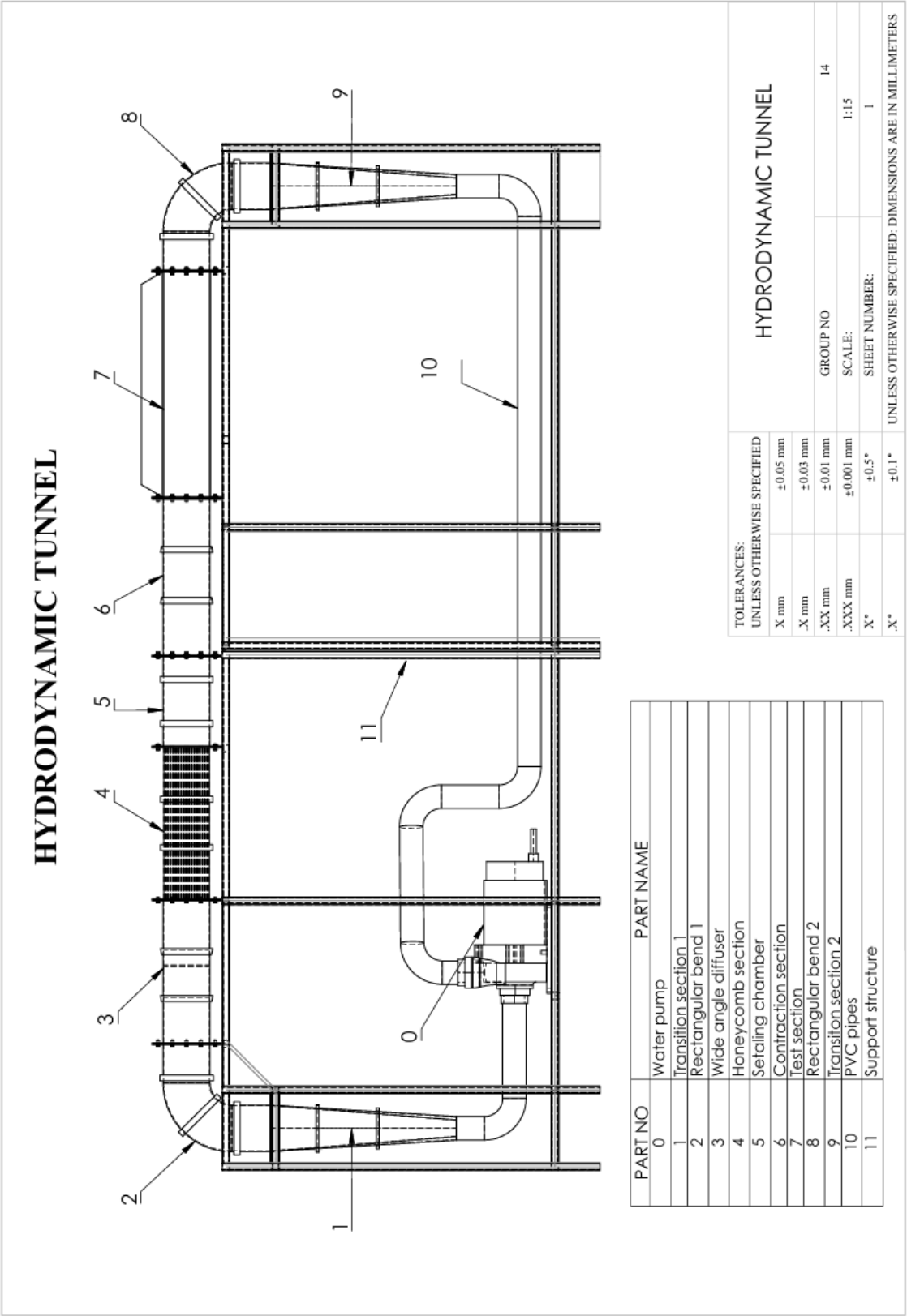
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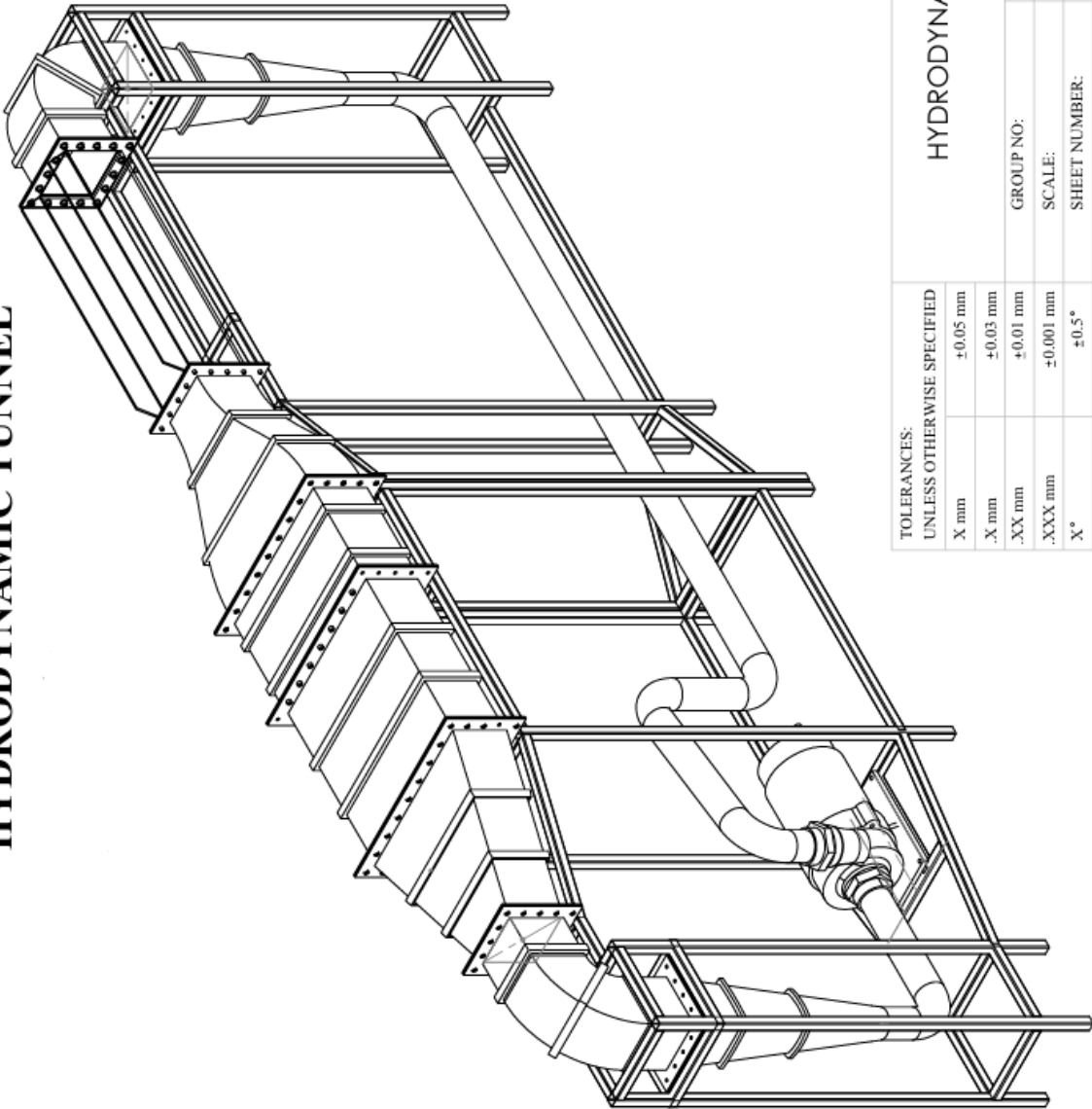
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8 Appendices

8.1 Appendix 1 – technical drawings

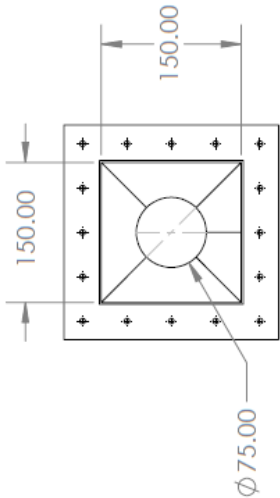


HYDRODYNAMIC TUNNEL

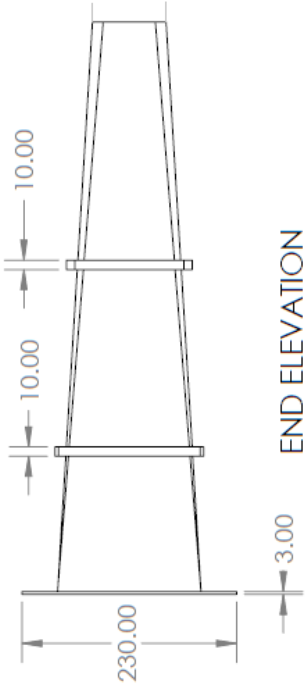


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.XXX mm	±0.001 mm	SCALE:		1:14
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.X°	±0.1°	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		

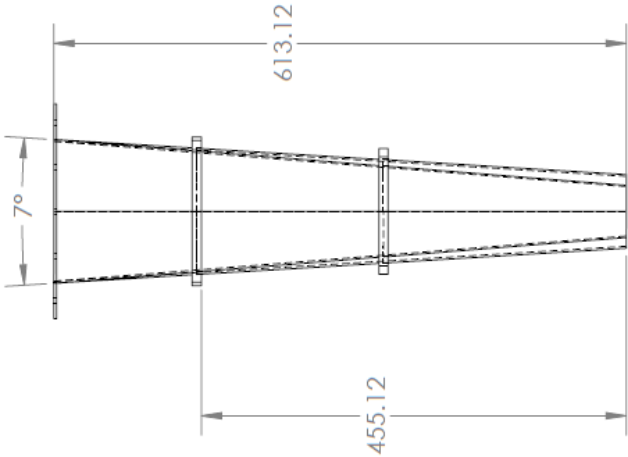
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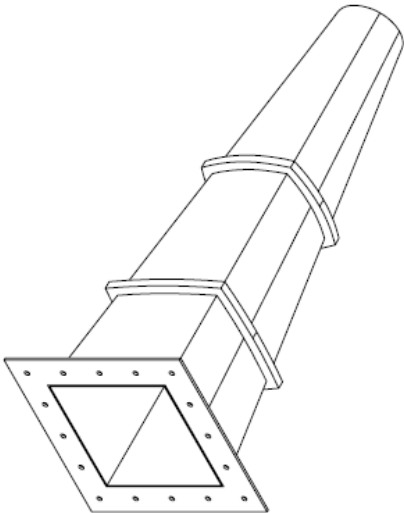
FRONT ELEVATION



END ELEVATION



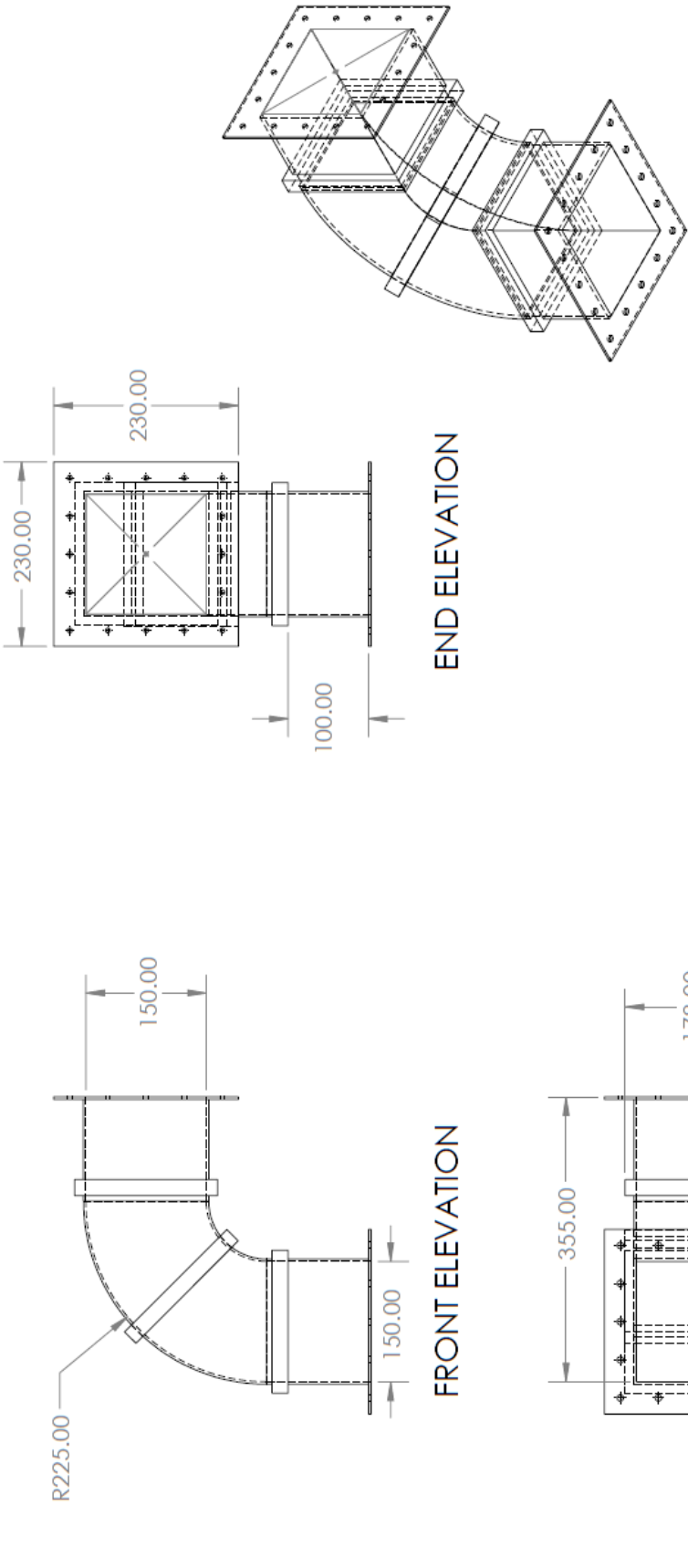
PLAN ELEVATION



ISOMETRIC VIEW

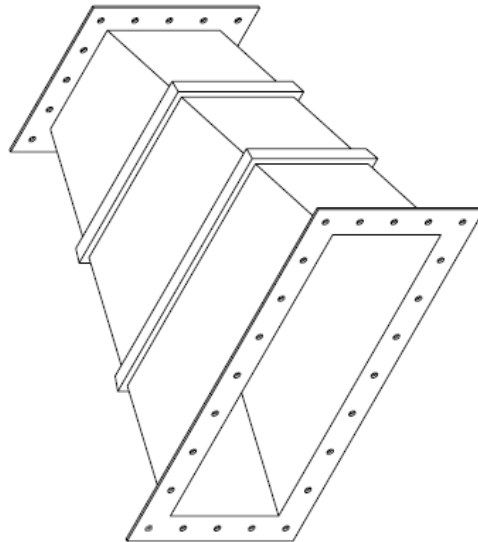
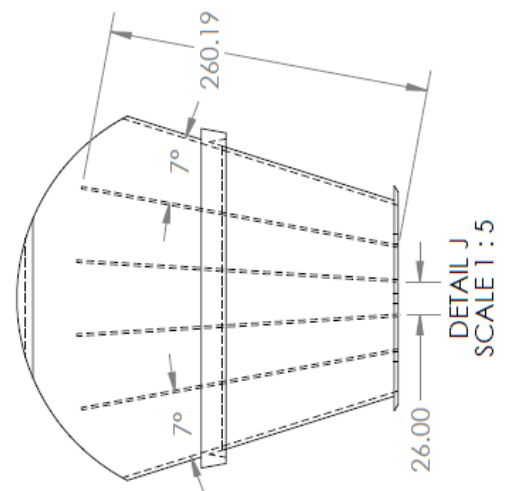
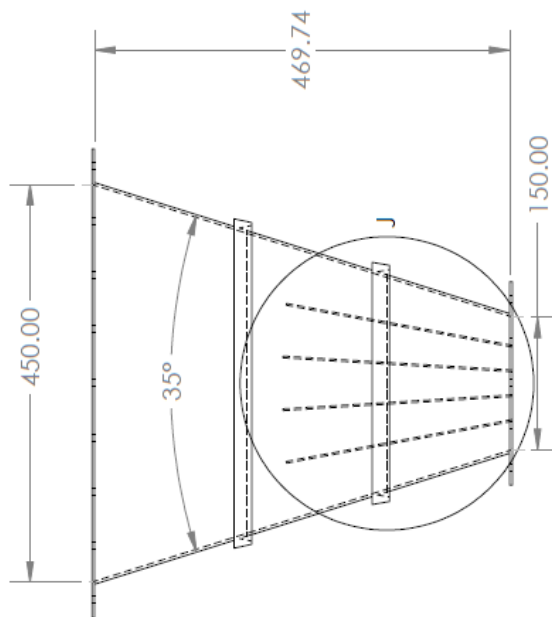
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RECTANGULAR BEND



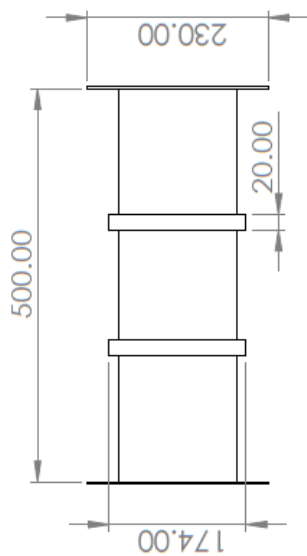
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WIDE ANGLE DIFFUSER

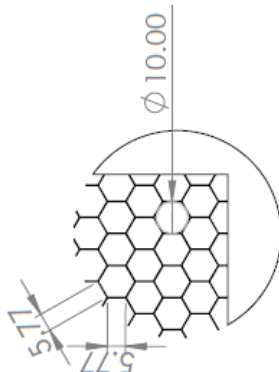


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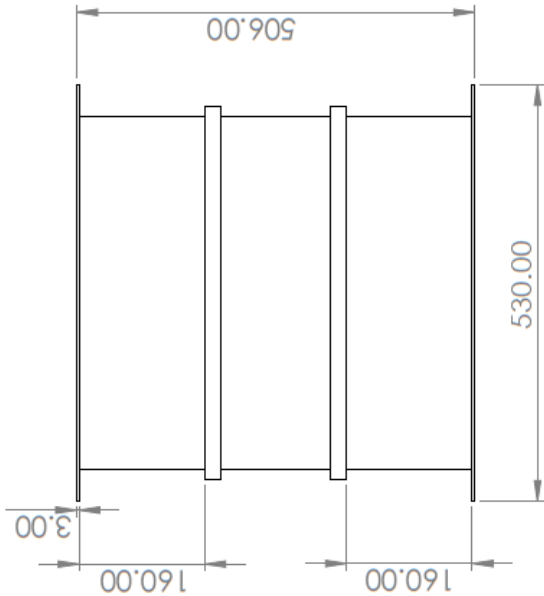
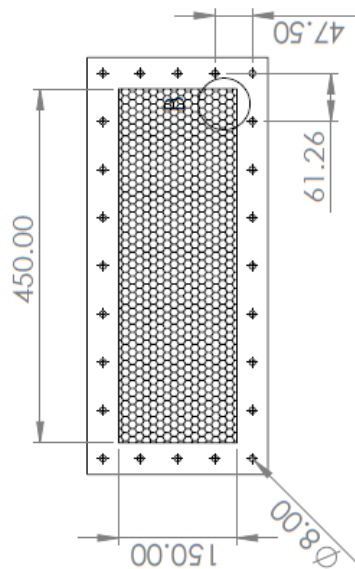
HONEYCOMB STRUCTURE



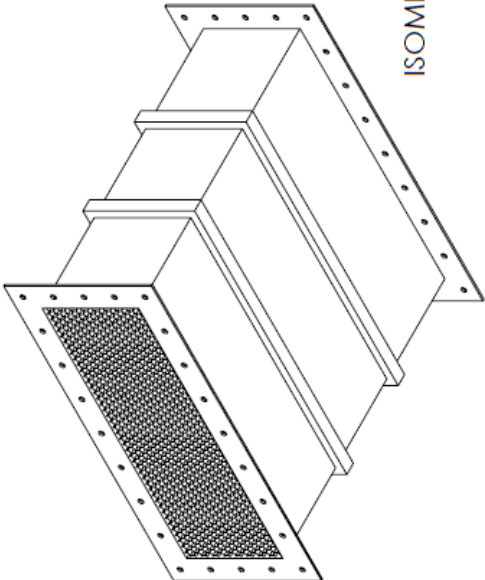
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FRONT ELEVATION



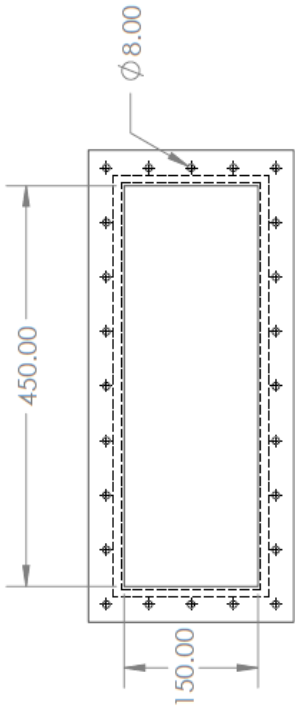
PLAN



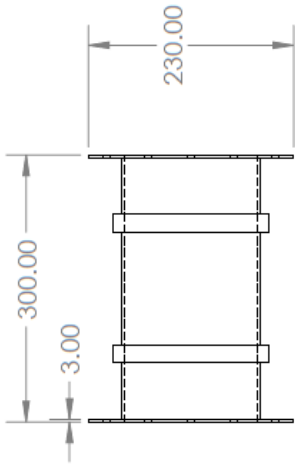
ISOMETRIC VIEW

TOLERANCES: UNLESS OTHERWISE SPECIFIED		HYDRODYNAMIC TUNNEL	
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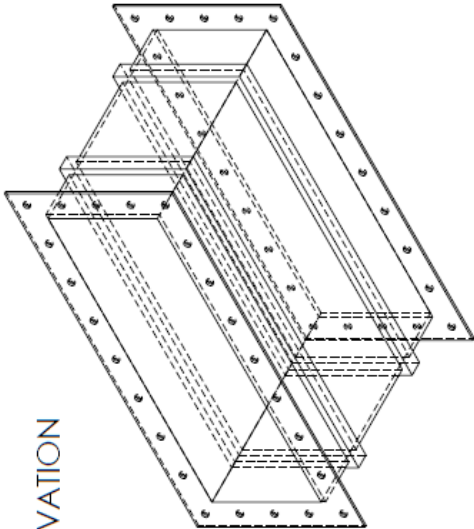
SETTLING CHAMBER



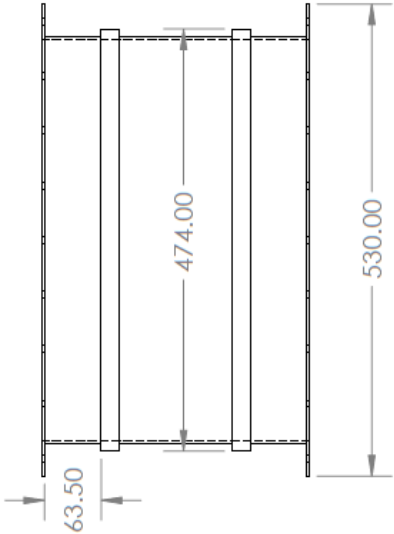
FRONT ELEVATION



END ELEVATION

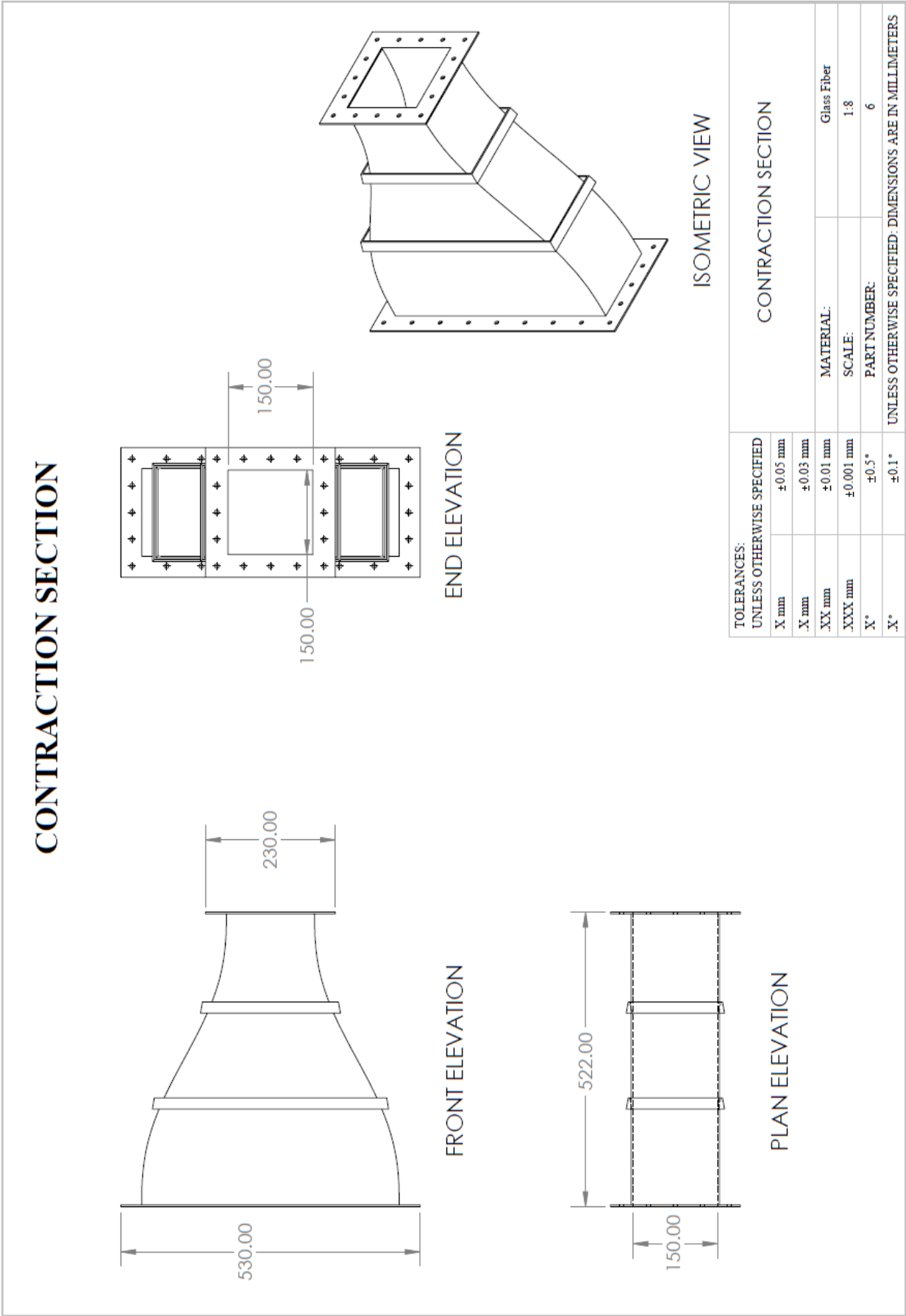


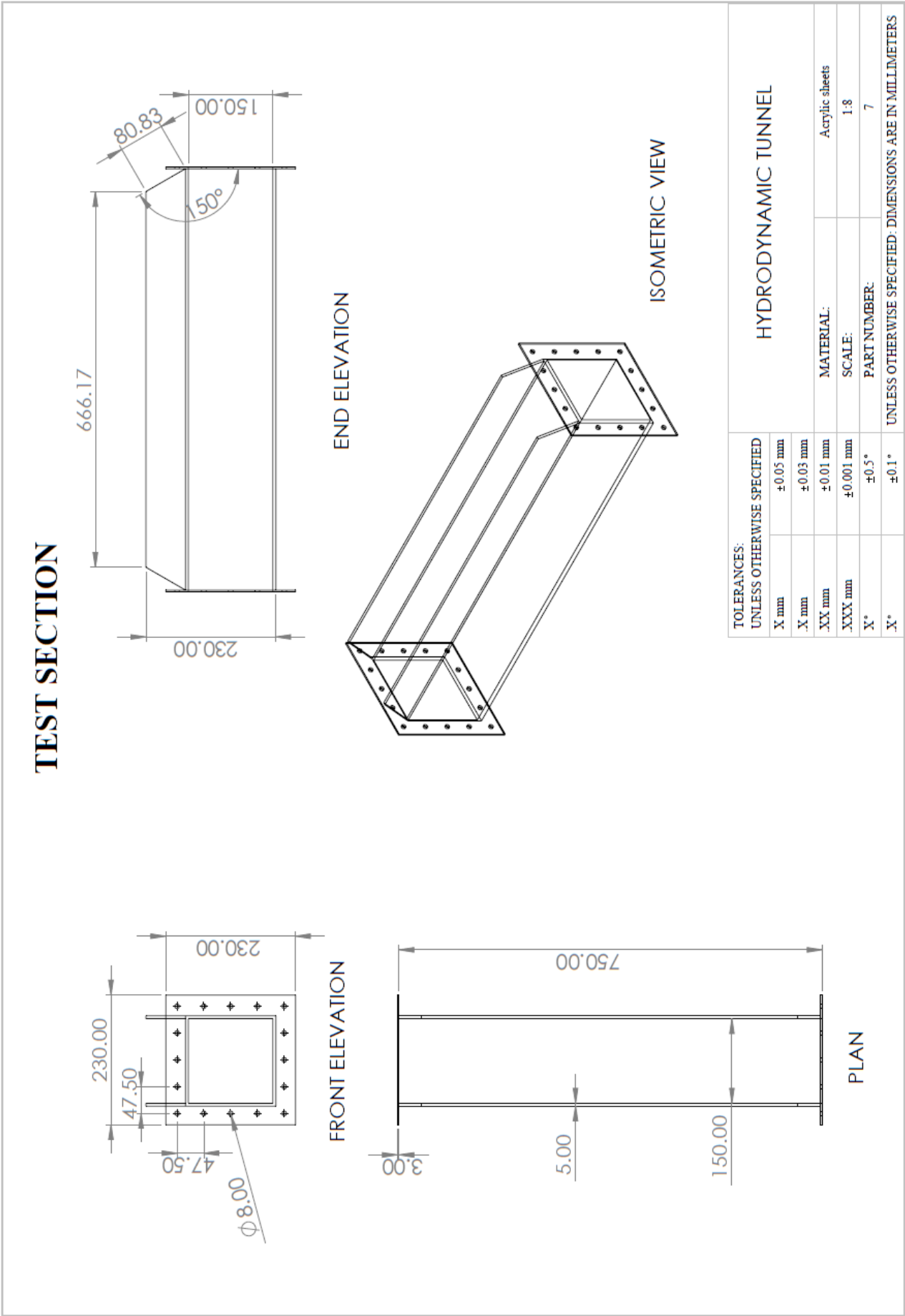
ISOMETRIC VIEW



PLAN ELEVATION

TOLERANCES: UNLESS OTHERWISE SPECIFIED		SETTLING CHAMBER		
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XX mm	±0.01 mm	MATERIAL:	Glass Fiber	
XXX mm	±0.001 mm	SCALE:	1:7	
X°	±0.5°	PART NUMBER:	5	
X°	±0.1°	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		





8.2 Appendix 2- Head loss calculation model

```

clc; clear;

% major head loss equation
function hL = majorLoss(f, L, D, V, g)
    hL = f * (L/D) * (V^2/(2*g));
end

% calculating velocity
function V = velocity(Q, D)
    A = pi*D^2/4;
    V = Q/A;
end

% calculating raynolds number
function Re = reynolds(rho, V, D, mu)
    Re = rho*V*D/mu;
end

% minor loss calculation for bends with bend radius of R diameter of D
function Kb = Kb_bend(R, D)

ratio = R / D;

if ratio >= 0 && ratio < 1
    Kb = 0.3;
elseif ratio >= 1 && ratio < 8
    Kb = 0.2;
elseif ratio >= 8 && ratio < 12
    Kb = 0.3;
elseif ratio >= 12 && ratio <= 20
    Kb = 0.4;
else
    warning('R/D ratio outside defined range. Using Kb = 0.4');
    Kb = 0.4;
end

end

function h_minor = contraction_minor_loss(Q, D1, D2, g)

% Diameter ratio
r = D2 / D1;

% Coefficient c (Merriman)
c = 0.582 + (0.0418 / (1.1 - r));

% Velocity in smaller pipe
Vs = Q / (pi * D2^2 / 4);

% Minor head loss due to contraction
h_minor = ((1/c) - 1)^2 * (Vs^2 / (2*g));

end

```

```

%{
% calculating friction factor for each component in turbulent region
colebrook = @(f) ...
    1/sqrt(f) + 2*log10( (eps/(3.7*d)) + (2.51/(Re*sqrt(f))) );

f = fzero(colebrook, 0.02);

disp(['Friction factor f = ', num2str(f)])
%}

%defining parameters
% if we need to change the motor parameters as well, we can uncomment this
%{
hPump = input('Enter pump head value (m): ');
Q = input('Enter flow rate (m^3/s): ');
rho = input('Enter fluid density (kg/m^3): ');
mu = input('Enter fluid dynamic viscosity (Pa.s): ');
g = 9.81; % acceleration due to gravity (m/s^2)
h0 = hPump;
%}

disp('=== HYDRODYNAMIC TUNNEL MAJOR HEAD LOSS CALCULATION ===')

g = 9.81; % gravity
h0 = 9.144; % meters -----
% ----- FLOW RATE -----

% Given flow rate in gallons per hour
Q_gph = input('Input the Gallons per hour value of the pump: ');

% Convert to cubic meters per second
Q = Q_gph * 0.00378541 / 3600;

% Number of components
nComp = 12;

%% List of components as in dineth's layout

% 0: pump
% 1: small straight pipe after pump
% 2: 90 degree bend
% 3: diffuser, also working as the reservoir from lower diameter to higher
diameter
% 4: 90 degree bend, may be with veins inside
% 5: diffuser from medium area to much higher area
% 6: Honeycomb structure
% 7: Contraction section
% 8: test section
% 9: 90 degree bend after the test section
% 10: diffuser from higher diameter to lower diameter
% 11: 90 degree bend
% 12: straight pump from bend to the pump

% Initialize head loss array
H = zeros(1, nComp);

```

```

%% ===== COMPONENT LOOP =====
for i = 1:nComp

    fprintf('\n--- Component %d ---\n', i)

    switch i

        % -----
        case 1 % Small straight pipe after pump
            disp('Small straight pipe')

            D = input('Pipe diameter D [m]: ');
            L = input('Pipe length L [m]: ');
            f = input('Friction factor f: ');

            V = velocity(Q, D);
            H(i) = f * (L/D) * (V^2/(2*g));

        % -----
        case {2,4,9,11} % 90-degree bends
            disp('90-degree bend')

            D = input('Pipe diameter D [m]: ');
            R = input('Bend radius R [m]: ');
            f = input('Friction factor f (for major loss): ');

            V = velocity(Q, D);

            % ----- Major loss (equivalent length) -----
            Leq = 0.5 * pi * R;
            h_major = f * (Leq/D) * (V^2/(2*g));

            % ----- Minor loss (Kb method) -----
            Kb = Kb_bend(R, D);
            h_minor = Kb * (V^2/(2*g));

            % ----- Total bend loss -----
            H(i) = h_major + h_minor;

            fprintf(' Major loss = %.4f m\n', h_major);
            fprintf(' Minor loss = %.4f m\n', h_minor);

        % -----
        case {3,5} % Diffusers (expansion)
            disp('Diffuser')

            D1 = input('Inlet diameter D1 [m]: ');
            D2 = input('Outlet diameter D2 [m]: ');
            L = input('Diffuser length L [m]: ');
            f = input('Equivalent friction factor f: ');
            D = (D1+D2)/2;

            V1 = velocity(Q, D);
            H(i) = f * (L/D) * (V1^2/(2*g));
    end
end

```



```

% -----
case 6 % Honeycomb
    disp('Honeycomb structure')

    D = input('Hydraulic diameter D [m]: ');
    L = input('Honeycomb length L [m]: ');
    %f = input('Equivalent friction factor f: ');
    V = velocity(Q, D);
    % Calculate Reynolds number for the honeycomb structure
    Re = reynolds(rho, V, D, mu);
    f = 64/Re;

    H(i) = f * (L/D) * (V^2/(2*g));

% -----
case 7 % Contraction

    disp('Contraction section')
    D1 = input('Inlet diameter D1 [m]: ');
    D2 = input('Outlet diameter D2 [m]: ');
    L = input('Contraction length L [m]: ');
    f = input('Equivalent friction factor f (major loss): ');

    % ----- MAJOR LOSS (Darcy, using average diameter) -----
    D_avg = (D1 + D2) / 2;
    V_avg = velocity(Q, D_avg);

    h_major = f * (L / D_avg) * (V_avg^2 / (2*g));

    % ----- MINOR LOSS (Merriman contraction loss) -----
    h_minor = contraction_minor_loss(Q, D1, D2, g);

    % ----- TOTAL LOSS FOR THIS COMPONENT -----
    H(i) = h_major + h_minor;

    fprintf(' Major loss (contraction) = %.4f m\n', h_major);
    fprintf(' Minor loss (contraction) = %.4f m\n', h_minor);

% -----
case 8 % Test section
    disp('Test section')

    D = input('Test section diameter D [m]: ');
    L = input('Test section length L [m]: ');
    f = input('Friction factor f: ');

    V = velocity(Q, D);
    % Calculate Reynolds number for the honeycomb structure
    Re = reynolds(rho, V, D, mu);
    f = 64/Re;
    H(i) = f * (L/D) * (V^2/(2*g));

% -----
case 10 % Diffuser (converging back)
    disp('Diffuser (high D to low D)')

```

```

D1 = input('Inlet diameter D1 [m]: ');
D2 = input('Outlet diameter D2 [m]: ');
L = input('Diffuser length L [m]: ');
f = input('Equivalent friction factor f: ');
D = (D1+D2)/2;

V1 = velocity(Q, D);
H(i) = f * (L/D) * (V1^2/(2*g));

% -----
case 12 % Straight pipe to pump
    disp('Straight pipe to pump')

    D = input('Pipe diameter D [m]: ');
    L = input('Pipe length L [m]: ');
    f = input('Friction factor f: ');

    V = velocity(Q, D);
    H(i) = f * (L/D) * (V^2/(2*g));

end

fprintf('Head loss H%d = %.4f m\n', i, H(i))

end

%% ===== RESULTS =====
disp(' ')
disp('=== COMPONENT HEAD LOSSES ===')

for i = 1:nComp
    fprintf('H%d = %.4f m\n', i, H(i))
end
disp('initial head: ',h0);
H_total = sum(H);
fprintf('\nTOTAL HEAD LOSS = %.4f m\n', H_total);

```